

MATSE: Operations Research

Lecture 2. The Simplex Method for Linear Programming

[based on slides by: Steven Kelk, Steve Chaplick]

Linear Programs: Basic Terminology

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■ **TABLE 3.1** Data for the Wyndor Glass Co. problem (goal: maximize profit)

Plant	Production Time per Batch, Hours		Production Time Available per Week, Hours
	Product		
	1	2	
1	1	0	4
2	0	2	12
3	3	2	18
Profit per batch	\$3,000	\$5,000	

$$\begin{aligned}
 &\text{Maximize } Z = 3x_1 + 5x_2, \\
 &\text{subject to } \begin{aligned} x_1 &\leq 4 \\ 2x_2 &\leq 12 \\ 3x_1 + 2x_2 &\leq 18 \end{aligned} \\
 &\text{and } x_1 \geq 0, x_2 \geq 0
 \end{aligned}$$

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decision
variables
 x_1, x_2

Maximize $Z = 3x_1 + 5x_2$,
subject to

$$x_1 \leq 4$$

$$2x_2 \leq 12$$

$$3x_1 + 2x_2 \leq 18$$

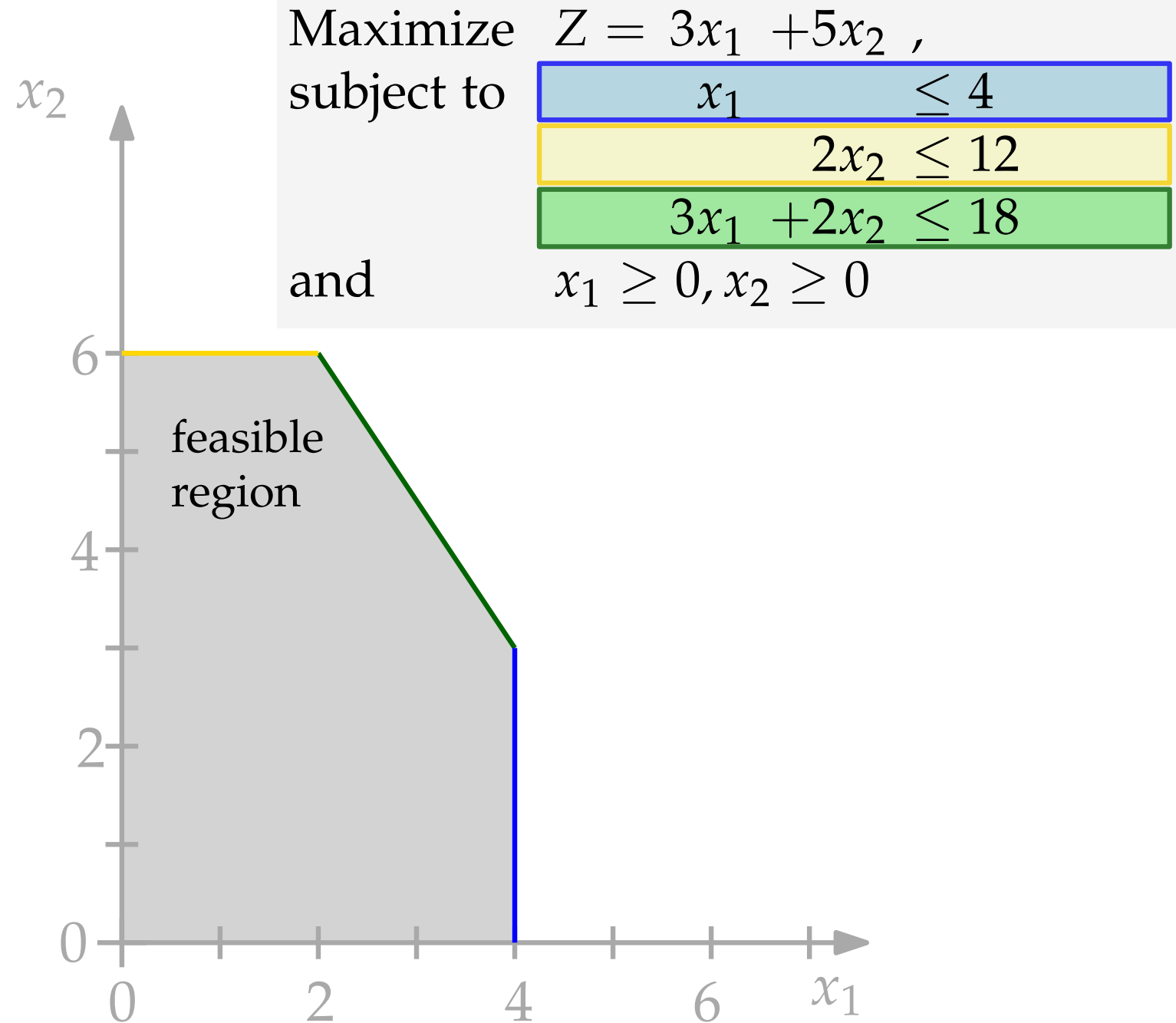
and

$$x_1 \geq 0, x_2 \geq 0$$

non-negativity constraints

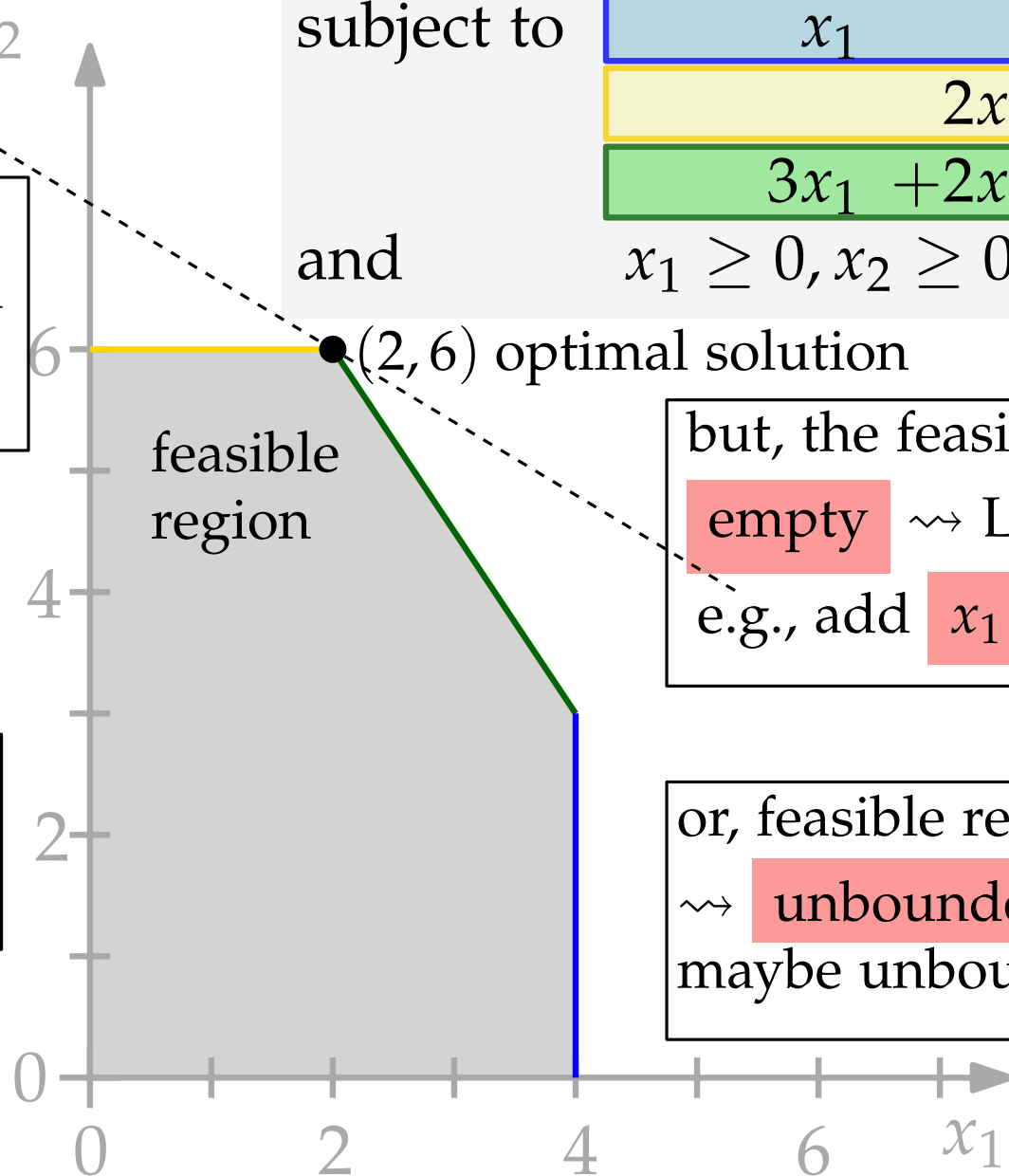
objective
function

functional
constraints



$$Z = 36 = 3x_1 + 5x_2$$

This feasible region is **non-empty**, and **bounded**



Maximize $Z = 3x_1 + 5x_2$,
subject to

$$x_1 \leq 4$$

$$2x_2 \leq 12$$

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$$x_1 \geq 0, x_2 \geq 0$$

and

(2, 6) optimal solution

feasible
region

but, the feasible region can be **empty** $\leadsto$ LP is **infeasible**
e.g., add $x_1 - x_2 \leq 4.5$

There can also be **many** optimal solutions!

or, feasible region can go to ∞
 $\leadsto$ **unbounded** LP, and maybe unbounded obj. value

Forms of LP

Given: $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$ [standard form]

where, for each entry b_i in b , $b_i \geq 0$, i.e., $b \geq 0$

Find: $x^* \in \mathbb{R}^n$ where $x^* = \arg \max \{c^T x \mid Ax \leq b, x \geq 0\}$

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This allows us to model more than you might expect,

- $\geq$ constraints $\rightarrow g(x) \geq b \rightarrow -g(x) \leq -b$
- $=$ constraints $\rightarrow g(x) = b \rightarrow g(x) \leq b$ and $-g(x) \leq -b$
- minimization $\rightarrow \text{Minimize } f(x) \rightarrow \text{Maximize } -f(x)$

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Why the standard form?

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when modelling a problem as an LP, use the standard form!

Some Linear Algebra

- Consider a set of m linear equalities in m variables.
- We express it as $Cx = d$, where C is an $m \times m$ matrix, x is a length- m column vector and d is a length- m column vector.
- What are necessary and sufficient conditions for this system of equalities to have a **unique solution**?

Some Linear Algebra

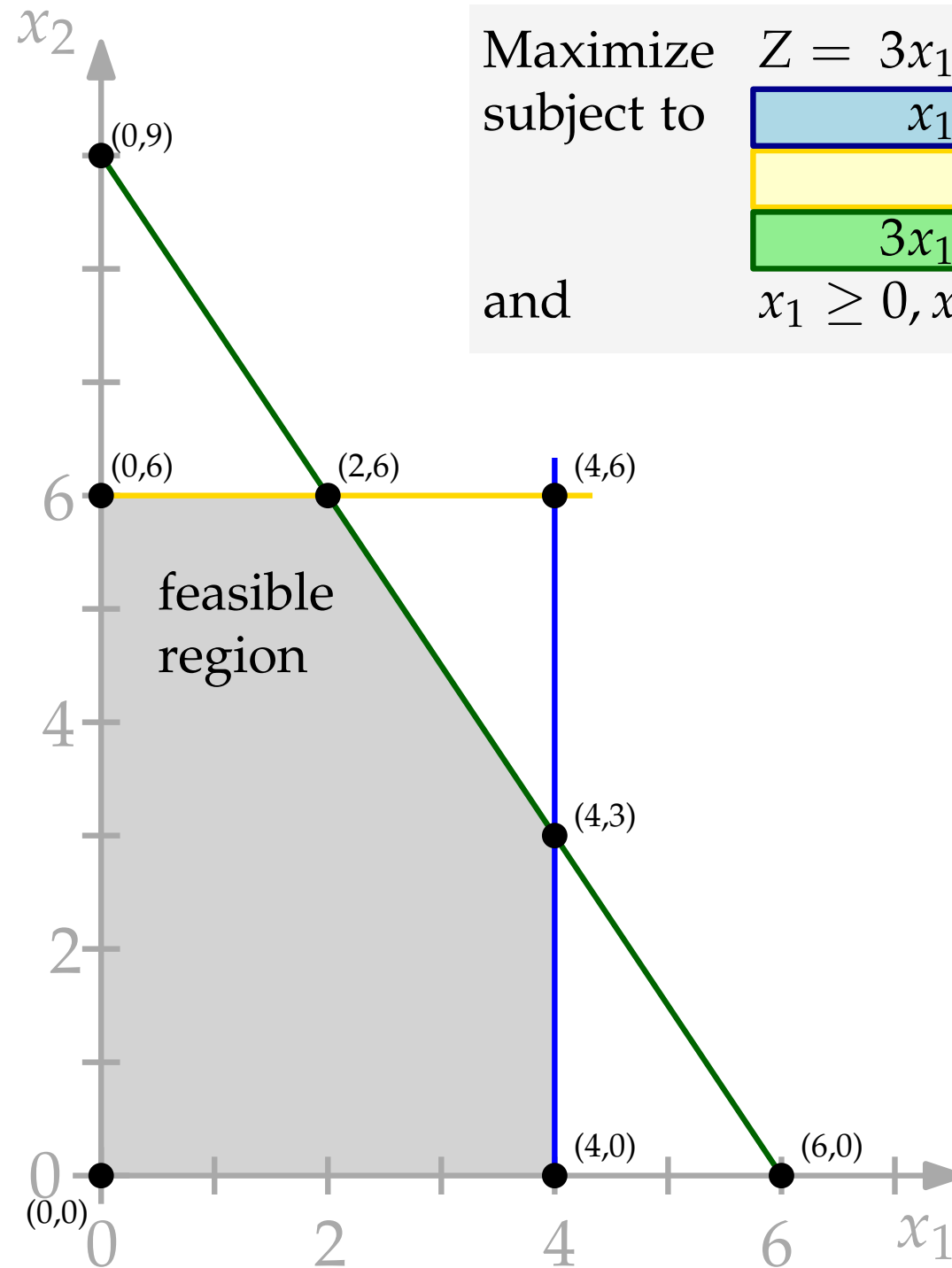
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 - $\text{row-rank}(C) = \text{column-rank}(C) = m$, equivalently
 - row set of C is **linearly independent**, equivalently,
 - cols. set of C is **linearly independent**, equivalently,
 - C is **invertible**, i.e., C^{-1} exists, i.e., $\det(C) \neq 0$.

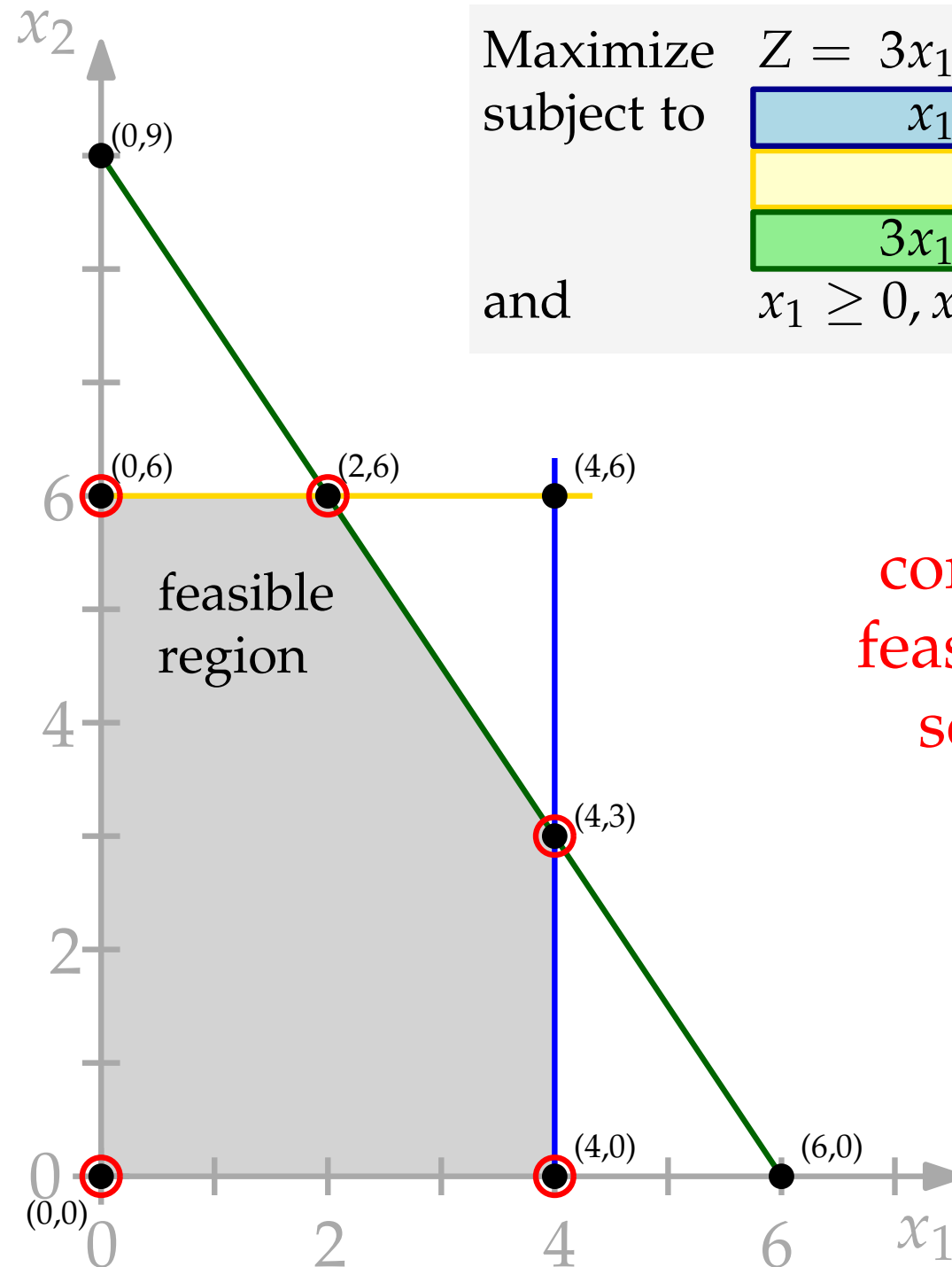
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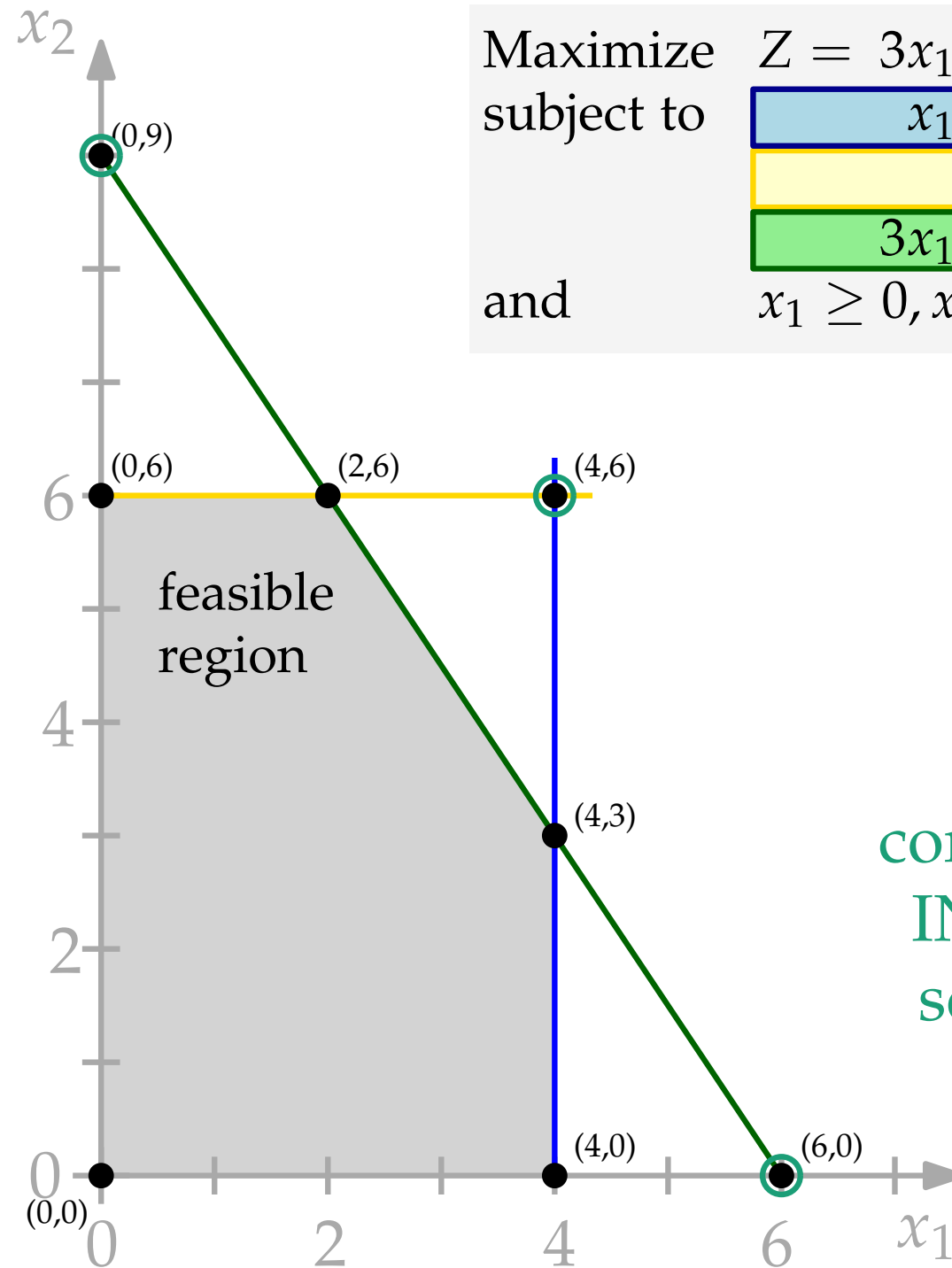
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- If these conditions fail, the system is either infeasible or infinitely many solutions exist.
- Simplex method iterates **corner-point feasible solutions**: certain unique solutions to *carefully* constructed systems of equalities (that are feasible for the linear program).





corner point
feasible (CPF)
solutions!



Maximize
subject to

$$Z = 3x_1 + 5x_2 ,$$

$$x_1 \leq 4$$

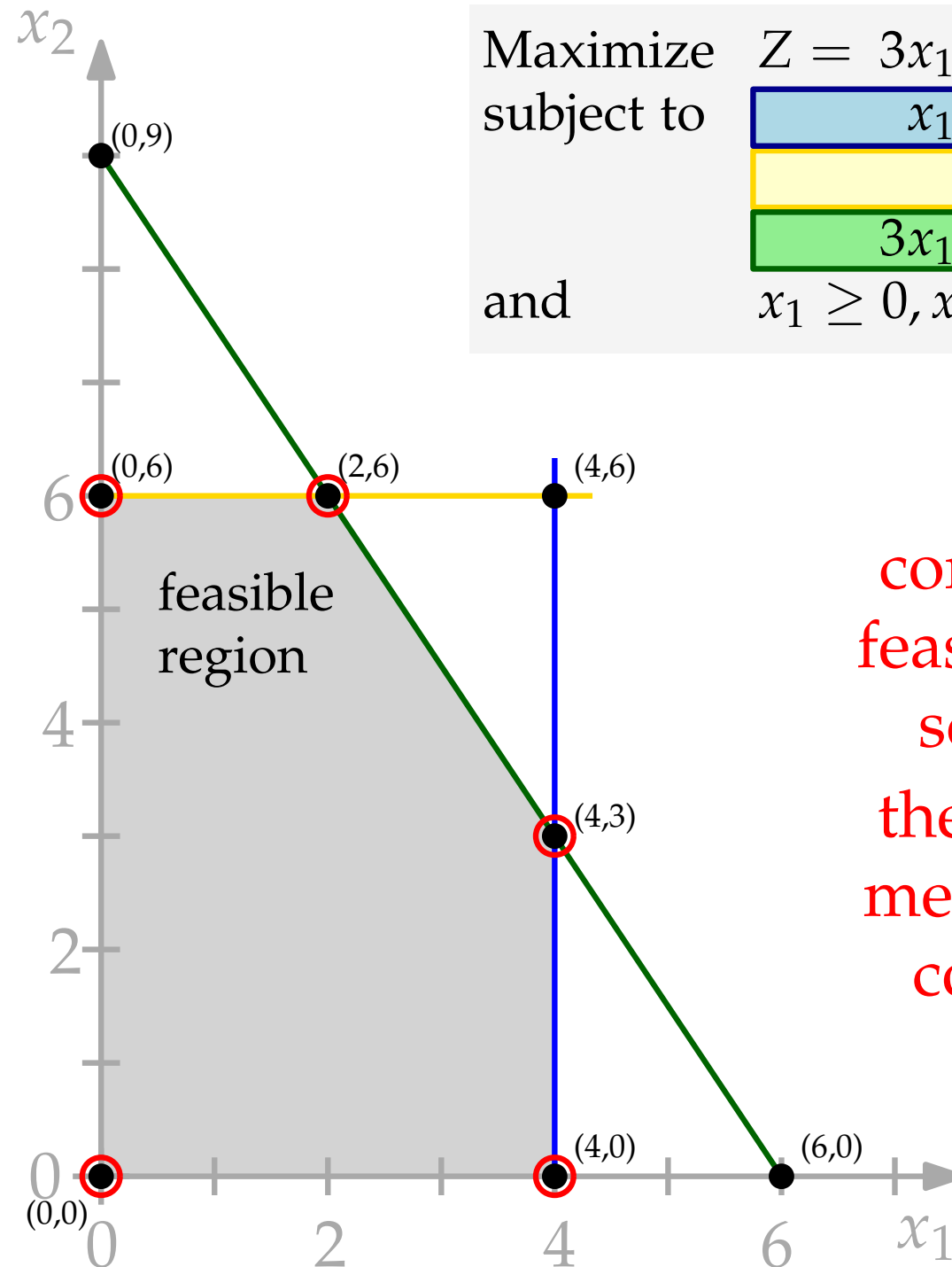
$$2x_2 \leq 12$$

$$3x_1 + 2x_2 \leq 18$$

and

$$x_1 \geq 0, x_2 \geq 0$$

corner-point
INfeasible
solutions!

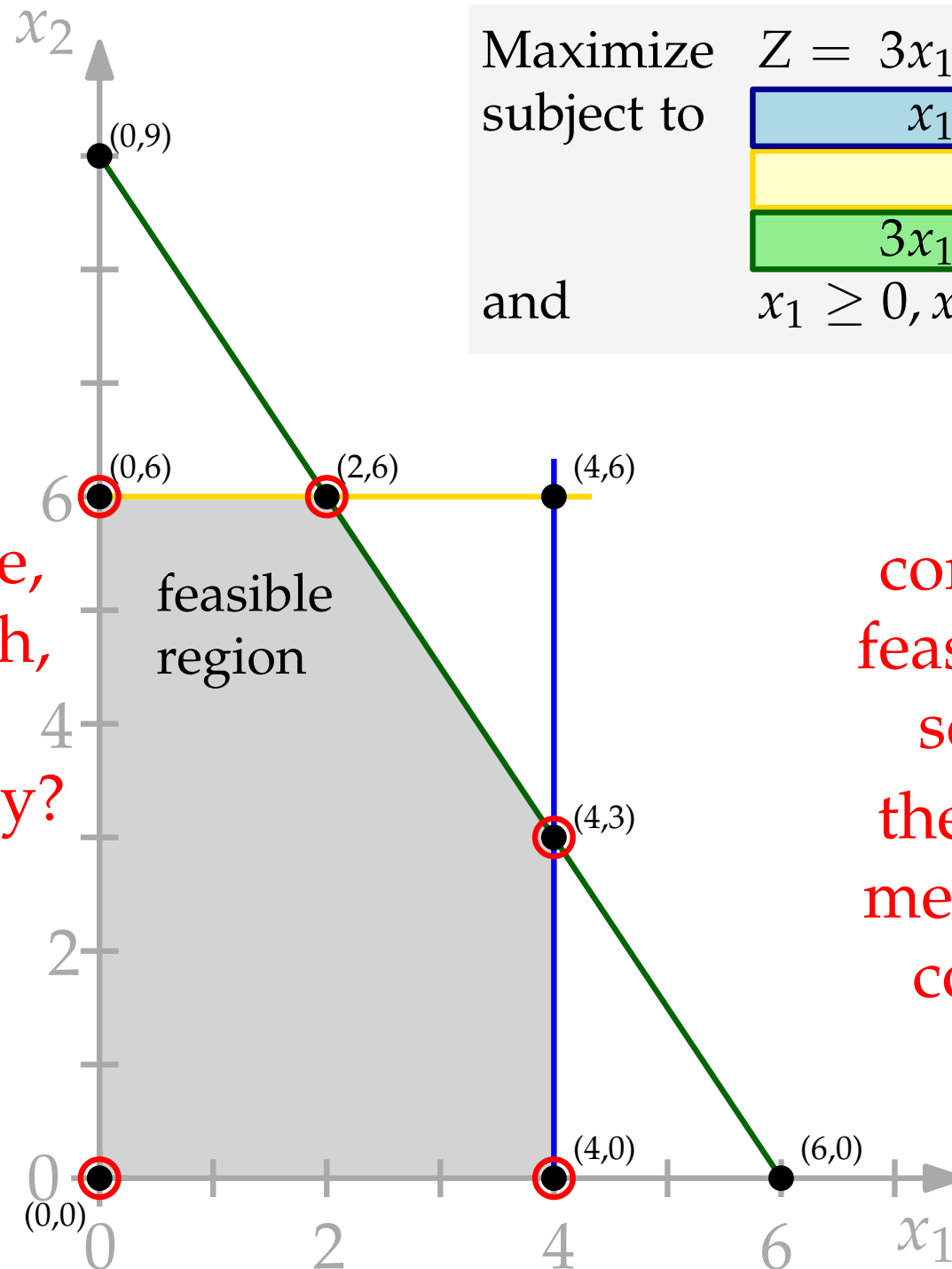


Maximize $Z = 3x_1 + 5x_2$,
 subject to

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and $x_1 \geq 0, x_2 \geq 0$

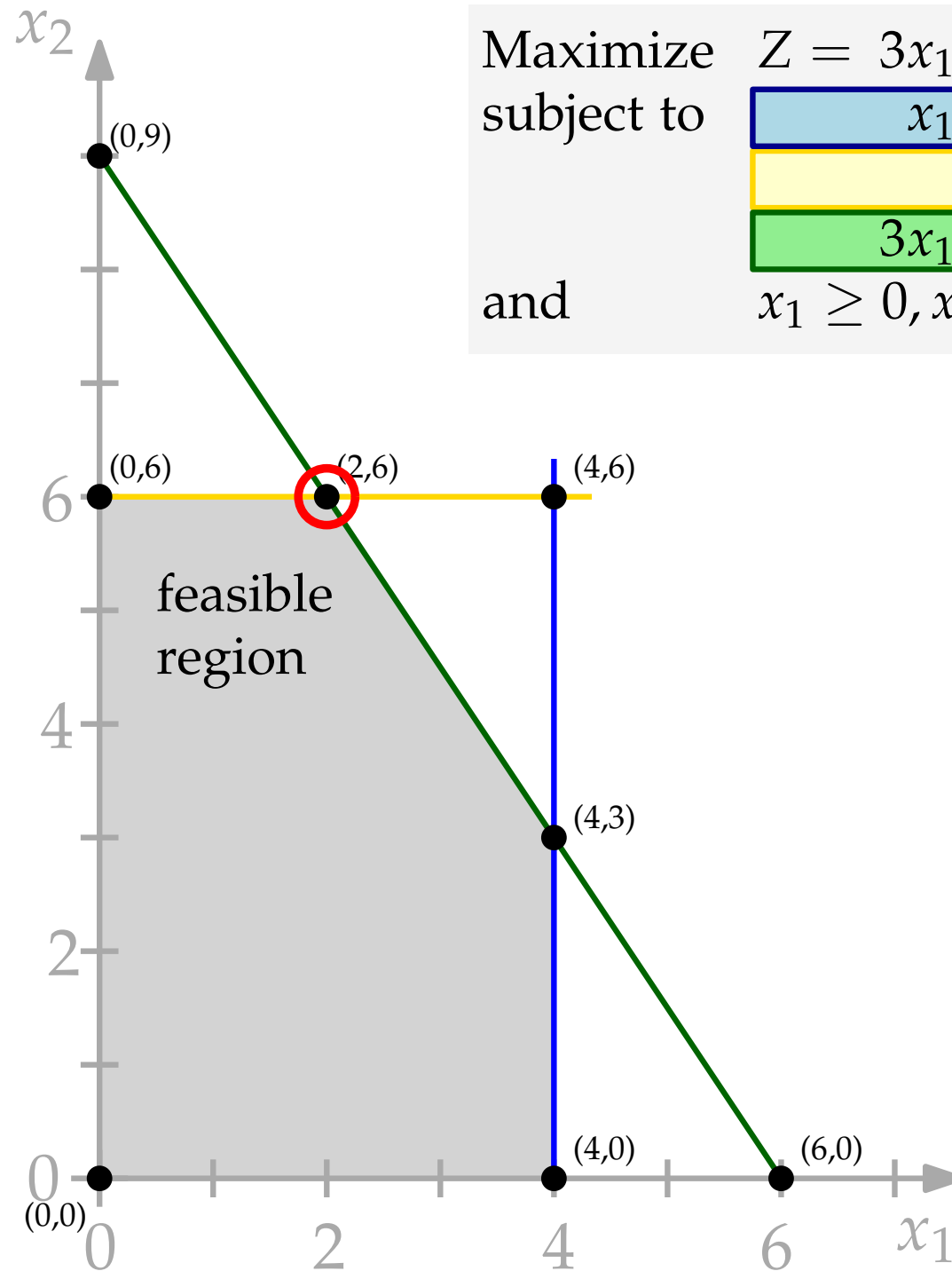
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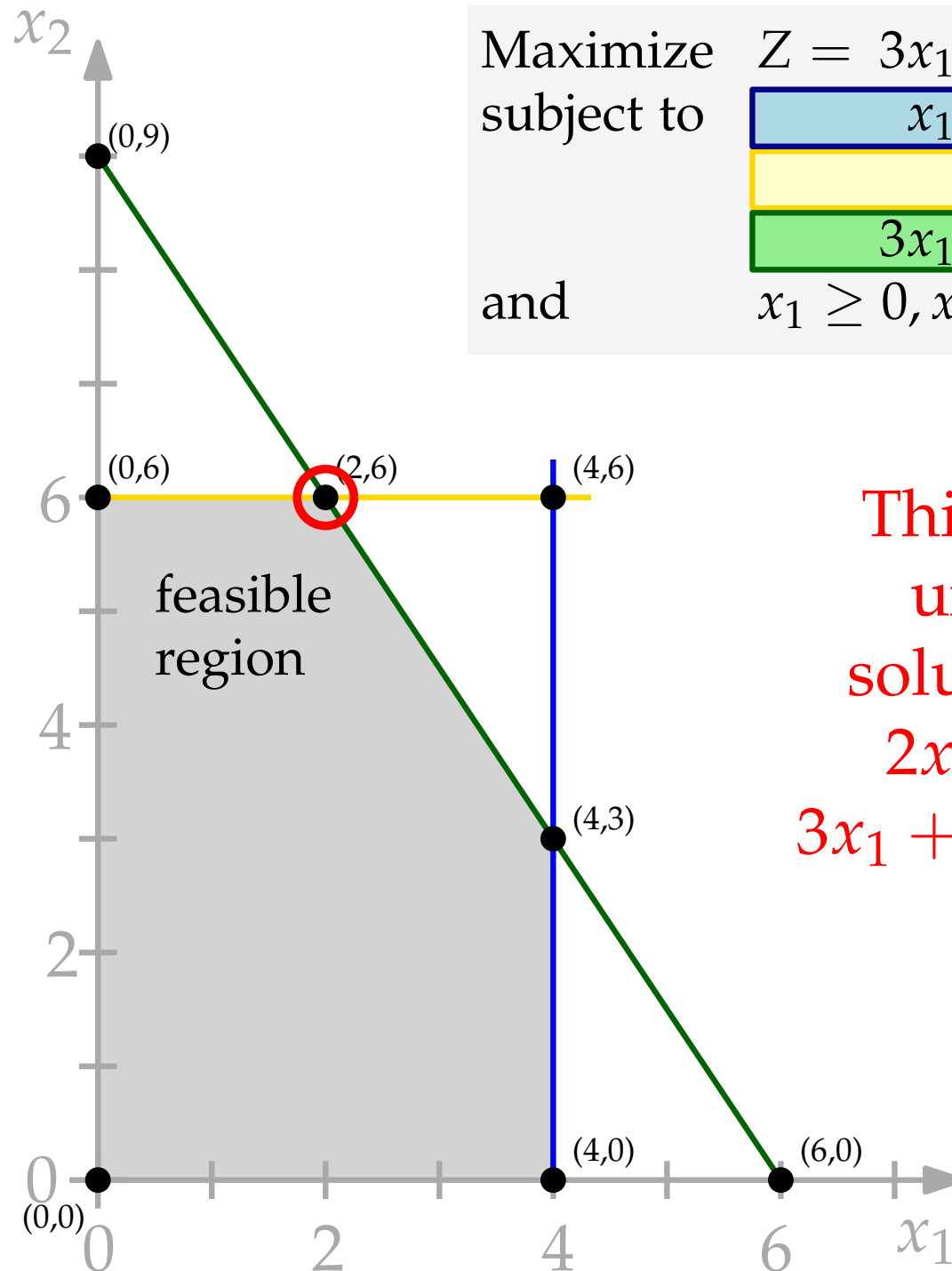


How to define,
and work with,
these CPF
mathematically?

corner point
feasible (CPF)
solutions!
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Consider the
CPF (2, 6)





Consider the
CPF $(2,6)$

This is the
unique
solution to:
 $2x_2 = 12$
 $3x_1 + 2x_2 = 18$

What's happening here?

- Cornerpoint feasible solutions have something to do with unique solutions to systems of **equalities** induced by subsets of the inequalities in the linear program (and the non-negativity constraints).

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- But linear programs are much more complex than systems of equalities. They are systems of **inequalities** (and we are looking for *optimal* solutions).

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- Cornerpoint feasible solutions have something to do with unique solutions to systems of **equalities** induced by subsets of the inequalities in the linear program (and the non-negativity constraints).
- But linear programs are much more complex than systems of equalities. They are systems of **inequalities** (and we are looking for *optimal* solutions).
- Solutions to systems of **inequalities** can be *projected* to solutions of systems of **equalities** in higher dimension (i.e. with more variables), and this is exactly what the Simplex method does.

Projecting “higher”

Consider the first constraint from our example:

$$x_1 \leq 4$$

where $x_1 \geq 0$

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Consider the first constraint from our example:

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$$\text{where } x_1 \geq 0$$

equivalent to

$$x_1 + x_3 = 4$$

$$\text{where } x_1, x_3 \geq 0$$

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slack variable: x_3
encodes the gap between
LHS and RHS

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When $x_3 = 0$, the inequality is satisfied with **equality**,
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What about the other constraints?

Projecting “higher”

original LP:

Maximize $Z = 3x_1 + 5x_2$

subject to

$$x_1 \leq 4$$

$$2x_2 \leq 12$$

$$3x_1 + 2x_2 \leq 18$$

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subject to

$$\begin{array}{rcl} x_1 & & \leq 4 \\ & 2x_2 & \leq 12 \\ 3x_1 + 2x_2 & \leq & 18 \end{array}$$

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augmented form of LP:

Maximize $Z = 3x_1 + 5x_2$
subject to

$$\begin{array}{rcl} x_1 & \boxed{+x_3} & = 4 \\ & 2x_2 & \boxed{+x_4} = 12 \\ 3x_1 + 2x_2 & \boxed{ +x_5} & = 18 \end{array}$$

and $x_1, x_2, \boxed{x_3, x_4, x_5} \geq 0$

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and $x_1, x_2, \boxed{x_3, x_4, x_5} \geq 0$

If the LP is in the form $Ax \leq b$ (and x is constrained to be ≥ 0), where A has m rows and n columns, then:

the augmented form will have m equalities in $m + n$ variables.

Matrix A : rank, bases

$$A = \begin{matrix} & x_1 & x_2 & x_3 & x_4 & x_5 \\ \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 \\ 3 & 2 & 0 & 0 & 1 \end{bmatrix} & b = \begin{bmatrix} 4 \\ 12 \\ 18 \end{bmatrix} \end{matrix}$$

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- A has full row rank (3), due to the 0-1 pattern (3×3 identity matrix) on the slack variables

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- Thus, there exist subsets of 3 columns which are linearly independent (because col-rank = row-rank).

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- A has full row rank (3), due to the 0-1 pattern (3×3 identity matrix) on the slack variables
- Thus, there exist subsets of 3 columns which are linearly independent (because col-rank = row-rank).
- These subsets are in **many-to-1** correspondence with corner-point solutions!

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 - Thus, there exist subsets of 3 columns which are linearly independent (because col-rank = row-rank).
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- (in this example, the subsets and corner points correspond 1-to-1)

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Ex. subset: $\{x_2, x_3, x_5\}$.

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The solution is **unique** (the induced 3×3 matrix is **invertible**)

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The solution is **unique** (the induced 3×3 matrix is **invertible**)

- $(x_1, x_2, x_3, x_4, x_5) = (0, 6, 4, 0, 6)$

Maximize
subject to

$$Z = 3x_1 + 5x_2 ,$$

$$x_1 \leq 4$$

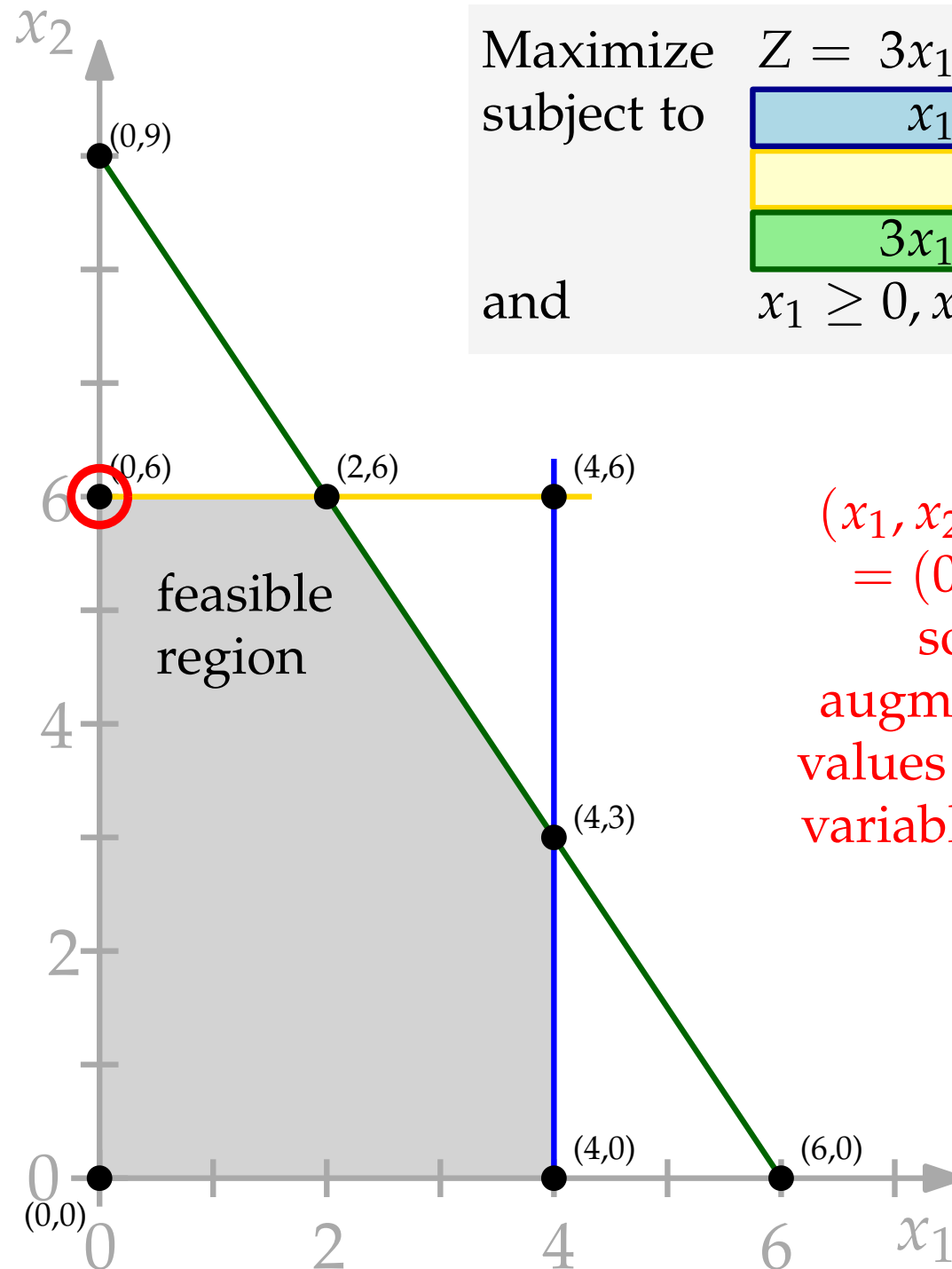
$$2x_2 \leq 12$$

$$3x_1 + 2x_2 \leq 18$$

and

$$x_1 \geq 0, x_2 \geq 0$$

$(0, 6, 4, 0, 6)$



$(x_1, x_2, x_3, x_4, x_5)$
 $= (0, 6, 4, 0, 6)$
 solution
 augmented with
 values of the slack
 variables x_3, x_4, x_5

Basic (B) vs. Non-Basic (NB)

$$A = \begin{matrix} & x_1 & x_2 & x_3 & x_4 & x_5 \\ \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 \\ 3 & 2 & 0 & 0 & 1 \end{bmatrix} & b = & \begin{bmatrix} 4 \\ 12 \\ 18 \end{bmatrix} \end{matrix}$$

$$(x_1, x_2, x_3, x_4, x_5) = (0, 6, 4, 0, 6)$$

Maximize $Z = 3x_1 + 5x_2$

subject to

$$x_1 + x_3 = 4$$

$$2x_2 + x_4 = 12$$

$$3x_1 + 2x_2 + x_5 = 18$$

and $x_1, x_2, x_3, x_4, x_5 \geq 0$

- This is a **basic solution** := augmented corner-point solution.

Basic (B) vs. Non-Basic (NB)

$$A = \begin{array}{ccccc} & x_1 & x_2 & x_3 & x_4 & x_5 \\ \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 \\ 3 & 2 & 0 & 0 & 1 \end{bmatrix} & & & & & \end{array} \quad b = \begin{bmatrix} 4 \\ 12 \\ 18 \end{bmatrix}$$

$$(x_1, x_2, x_3, x_4, x_5) = (0, 6, 4, 0, 6)$$

Maximize $Z = 3x_1 + 5x_2$

subject to

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$$\begin{array}{rrrrrcl} x_1 & & & +x_3 & & = & 4 \\ & 2x_2 & & & +x_4 & = & 12 \\ 3x_1 & +2x_2 & & & & +x_5 & = & 18 \end{array}$$

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$$A = \begin{array}{c} \text{NB} \quad \text{B} \quad \text{B} \quad \text{NB} \quad \text{B} \\ \begin{array}{ccccc} x_1 & x_2 & x_3 & x_4 & x_5 \\ \left[\begin{array}{ccccc} 1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 \\ 3 & 2 & 0 & 0 & 1 \end{array} \right] \end{array} \end{array} \quad b = \begin{bmatrix} 4 \\ 12 \\ 18 \end{bmatrix}$$

$$(x_1, x_2, x_3, x_4, x_5) = (0, 6, 4, 0, 6)$$

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- The basis here is $\{x_2, x_3, x_5\}$ and x_2, x_3, x_5 are **basic variables**.
- The variables x_1 and x_4 are **nonbasic variables**.

Basic (B) vs. Non-Basic (NB)

$$A = \begin{array}{c} \text{NB} \quad \text{B} \quad \text{B} \quad \text{NB} \quad \text{B} \\ \begin{array}{c} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{array} \\ \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 \\ 3 & 2 & 0 & 0 & 1 \end{bmatrix} \end{array} \quad b = \begin{bmatrix} 4 \\ 12 \\ 18 \end{bmatrix}$$

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and $x_1, x_2, x_3, x_4, x_5 \geq 0$

- Every **linearly independent subset of 3 columns is a basis**, and generates a **basic solution**, but not all will be feasible.

Basic (B) vs. Non-Basic (NB)

$$\begin{array}{ccccc}
 \text{NB} & \text{B} & \text{B} & \text{NB} & \text{B} \\
 x_1 & x_2 & x_3 & x_4 & x_5 \\
 A = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 \\ 3 & 2 & 0 & 0 & 1 \end{bmatrix} & b = \begin{bmatrix} 4 \\ 12 \\ 18 \end{bmatrix} \\
 (x_1, x_2, x_3, x_4, x_5) = (0, 6, 4, 0, 6)
 \end{array}$$

Maximize $Z = 3x_1 + 5x_2$

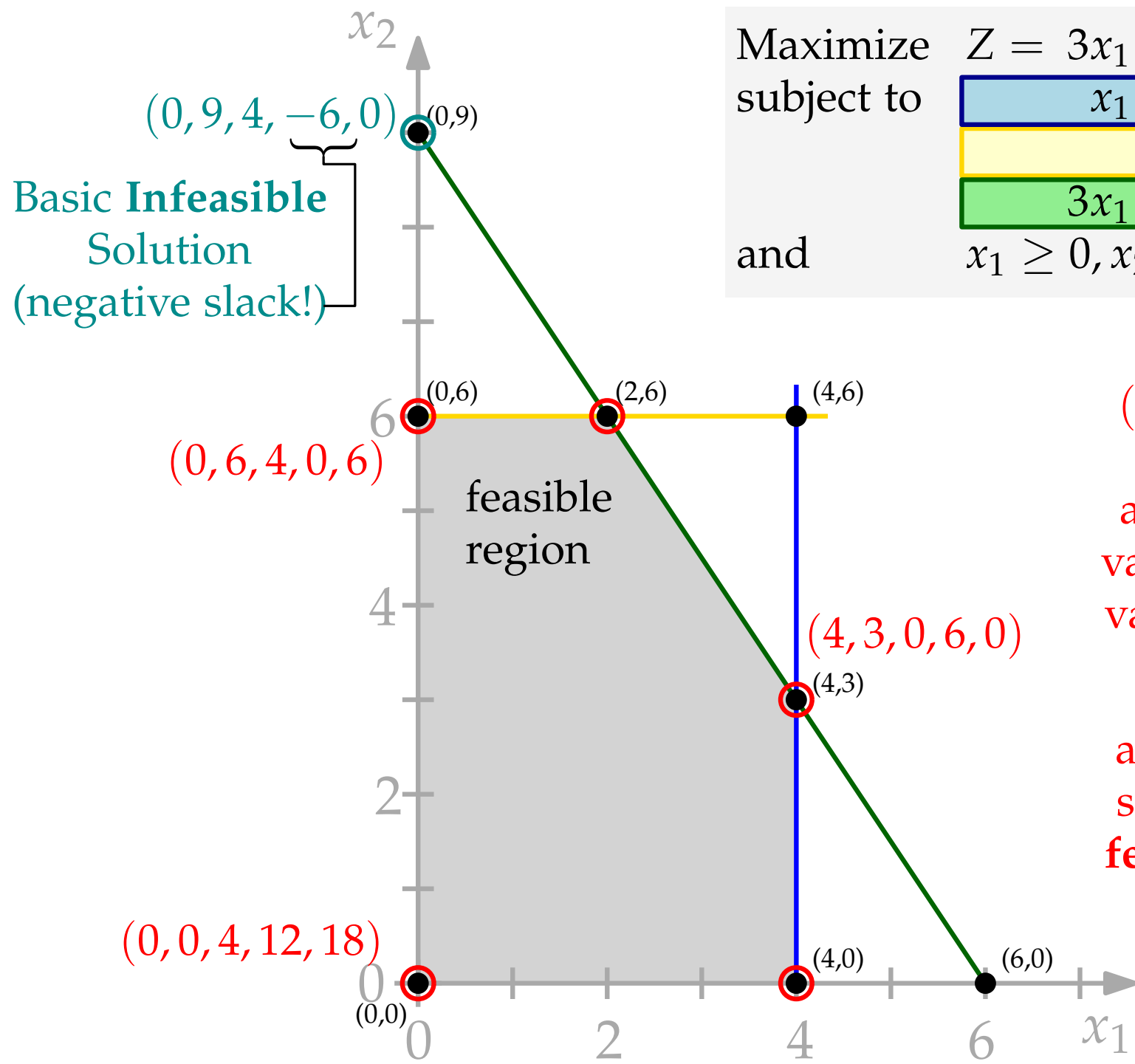
subject to

$$\begin{array}{rclcl}
 x_1 & & +x_3 & & = 4 \\
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 \end{array}$$

and $x_1, x_2, x_3, x_4, x_5 \geq 0$

- Every **linearly independent subset of 3 columns is a basis**, and generates a **basic solution**, but not all will be feasible.
- A basic solution is **feasible**, if and only if all the decision variables and slack variables are **non-negative**.

Maximize $Z = 3x_1 + 5x_2$,
 subject to $x_1 \leq 4$
 $2x_2 \leq 12$
 $3x_1 + 2x_2 \leq 18$
 and $x_1 \geq 0, x_2 \geq 0$



$(x_1, x_2, x_3, x_4, x_5)$
 = solution
 augmented with
 values of the slack
 variables x_3, x_4, x_5

augmented CPF
 solution = **basic
 feasible solution**
 (BFS)

Summary so far

- LP with n decision variables and m functional constraints $\rightsquigarrow$ augmented form of the LP is a set of m equalities in $(n + m)$ variables i.e. an **underdetermined system of equalities**.

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- LP with n decision variables and m functional constraints $\rightsquigarrow$ augmented form of the LP is a set of m equalities in $(n + m)$ variables i.e. an **underdetermined system of equalities**.
- To obtain a unique solution to this system of equalities, we **force** n of the variables to zero (such that the remaining columns are linearly independent)

Summary so far

- Geometric interpretation:
this corresponds to the choice of n constraint boundaries which we ask to be simultaneously satisfied with equality.
(Constraint boundaries: functional constraints & ≥ 0 constraints).

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Note: you can view this geometric interpretation as an alternative to the algebraic definition from the earlier slides - but they are equivalent! The algebraic definition places more emphasis on basic variables, the geometric definition places more emphasis on non-basic variables.

Check Table 4.2 from the book!

Summary so far

- The n variables that we *choose* to force to zero, are known as the **nonbasic** variables.

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- **The idea** : *choose* n variables to be **nonbasic**, the basic variables (i.e. the basis) is the set of m variables that remain, and their values are the **unique solution** to the system of equalities obtained after driving the nonbasic variables to zero.

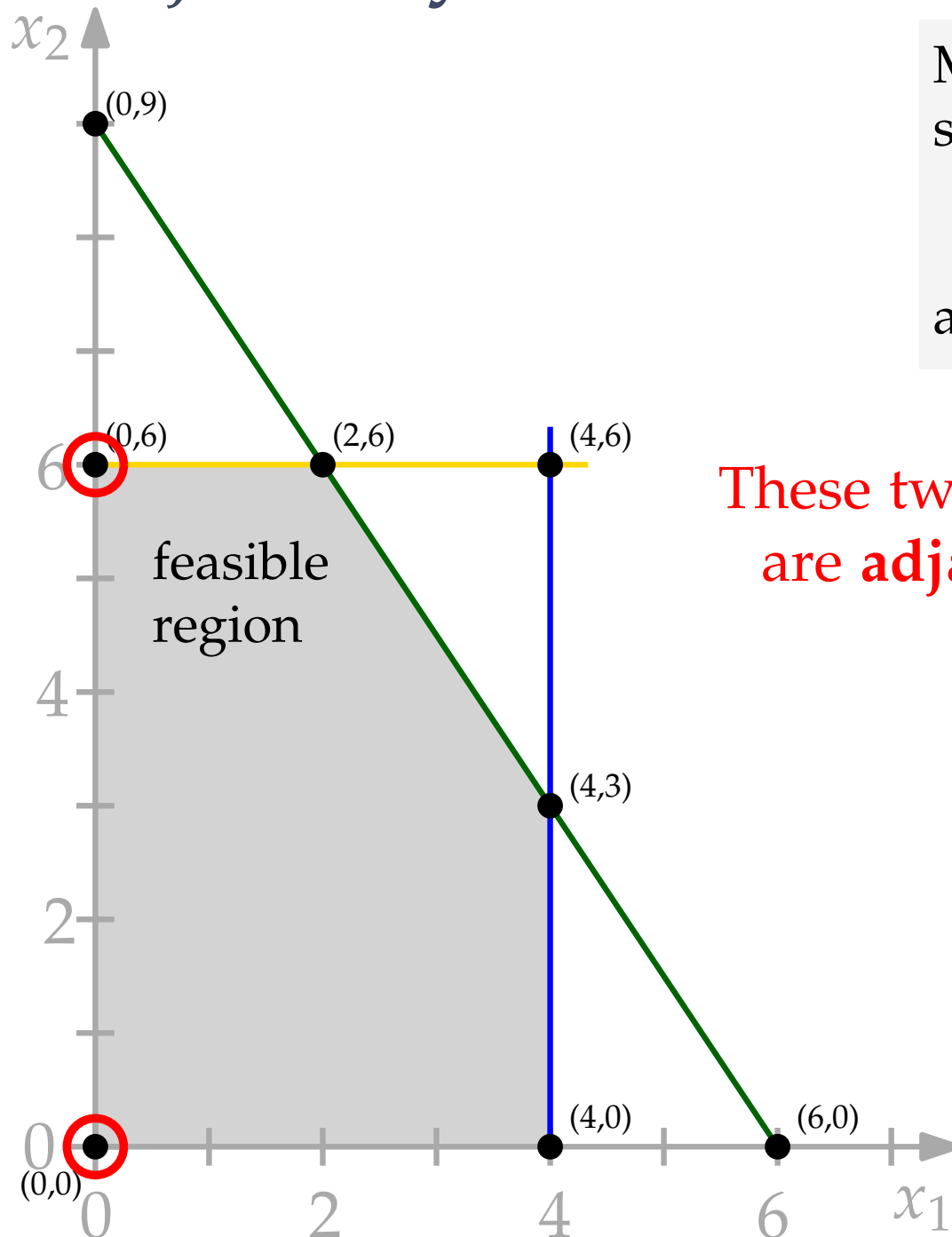
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⇒ (1) nonbasic means **zero**, but (2) basic **can** also be **zero**

Adjacency

Adjacency



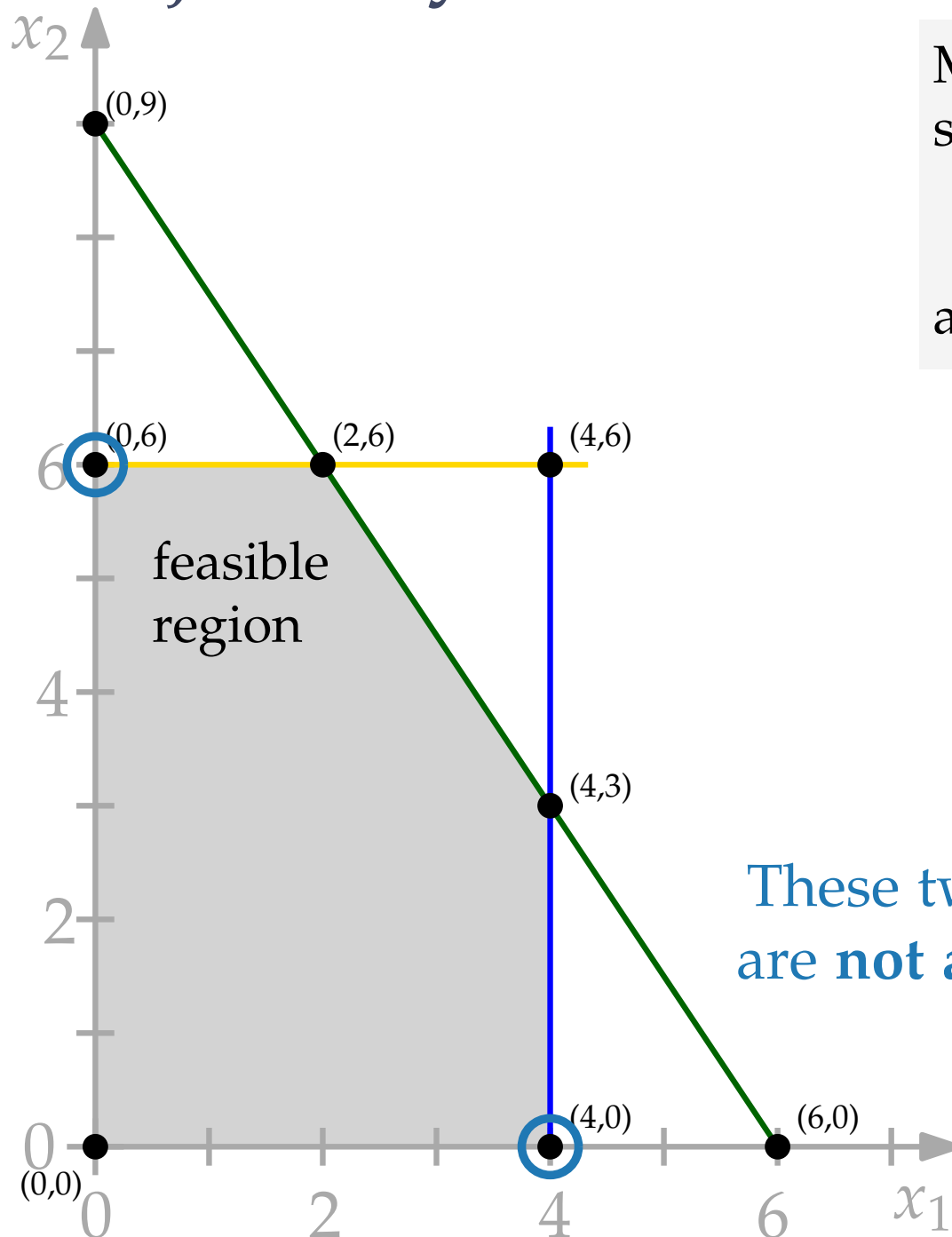
These two BFSs
are adjacent

Maximize $Z = 3x_1 + 5x_2$,
 subject to

$x_1 \leq 4$
$2x_2 \leq 12$
$3x_1 + 2x_2 \leq 18$

and $x_1 \geq 0, x_2 \geq 0$

Adjacency



Maximize
subject to

$$Z = 3x_1 + 5x_2 ,$$

$$x_1 \leq 4$$

$$2x_2 \leq 12$$

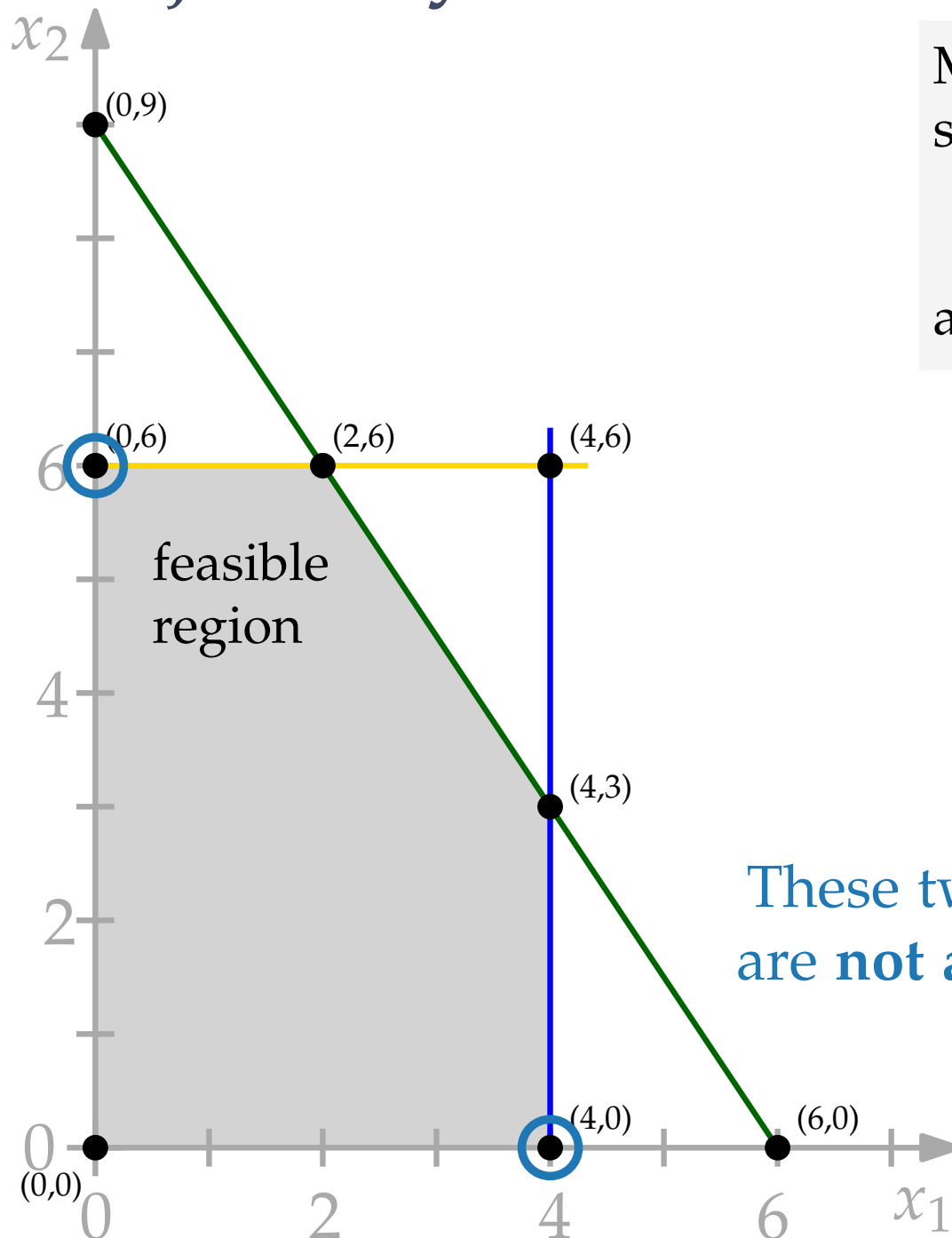
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These two BFSs
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Adjacency



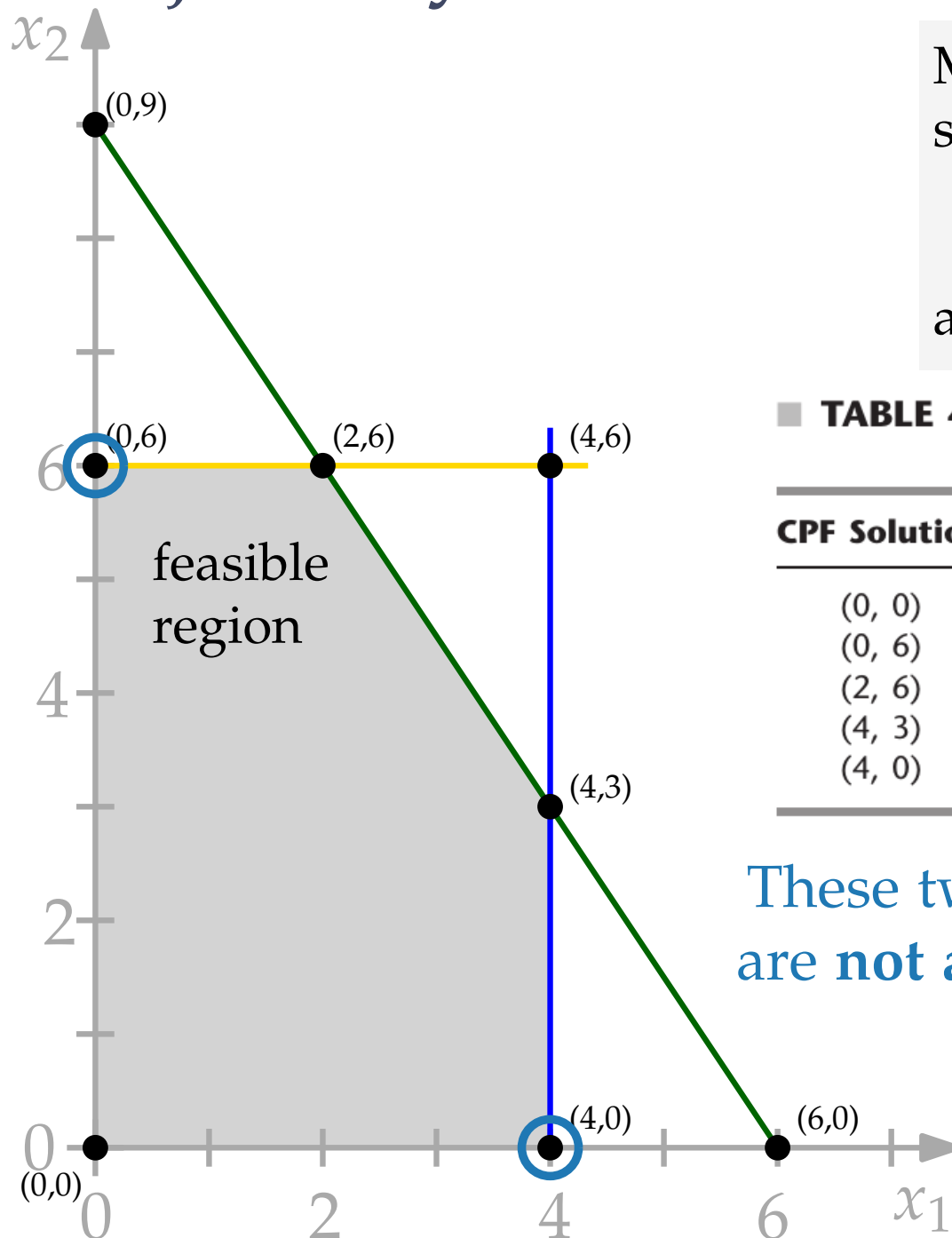
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Why? How do we formalize adjacency?

Adjacency



Maximize $Z = 3x_1 + 5x_2$,

subject to

$$x_1 \leq 4$$

$$2x_2 \leq 12$$

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and

$$x_1 \geq 0, x_2 \geq 0$$

TABLE 4.1 Adjacent CPF solutions for each CPF solution of the Wyndor Glass Co. problem

CPF Solution	Its Adjacent CPF Solutions
$(0, 0)$	$(0, 6)$ and $(4, 0)$
$(0, 6)$	$(2, 6)$ and $(0, 0)$
$(2, 6)$	$(4, 3)$ and $(0, 6)$
$(4, 3)$	$(4, 0)$ and $(2, 6)$
$(4, 0)$	$(0, 0)$ and $(4, 3)$

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Adjacency (special case)

In n dimensions, suppose at each CPF solution exactly n constraint boundaries meet

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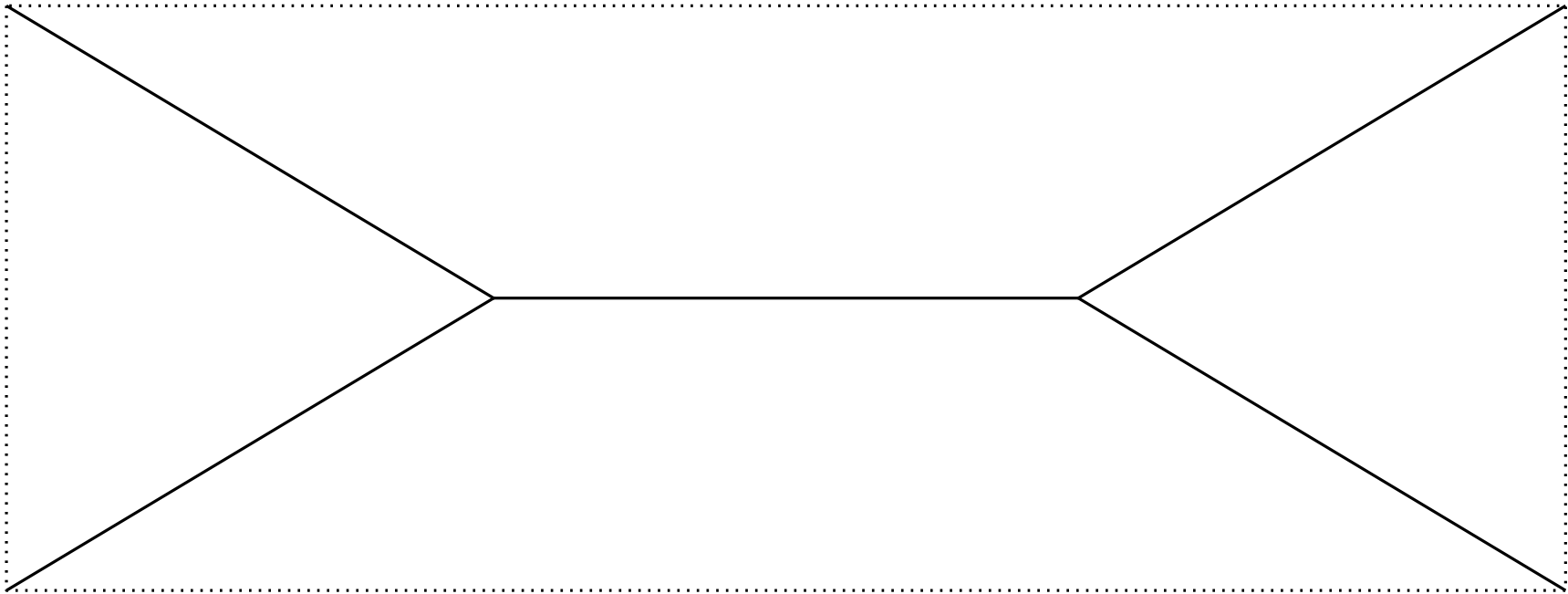
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Adjacency (special case)

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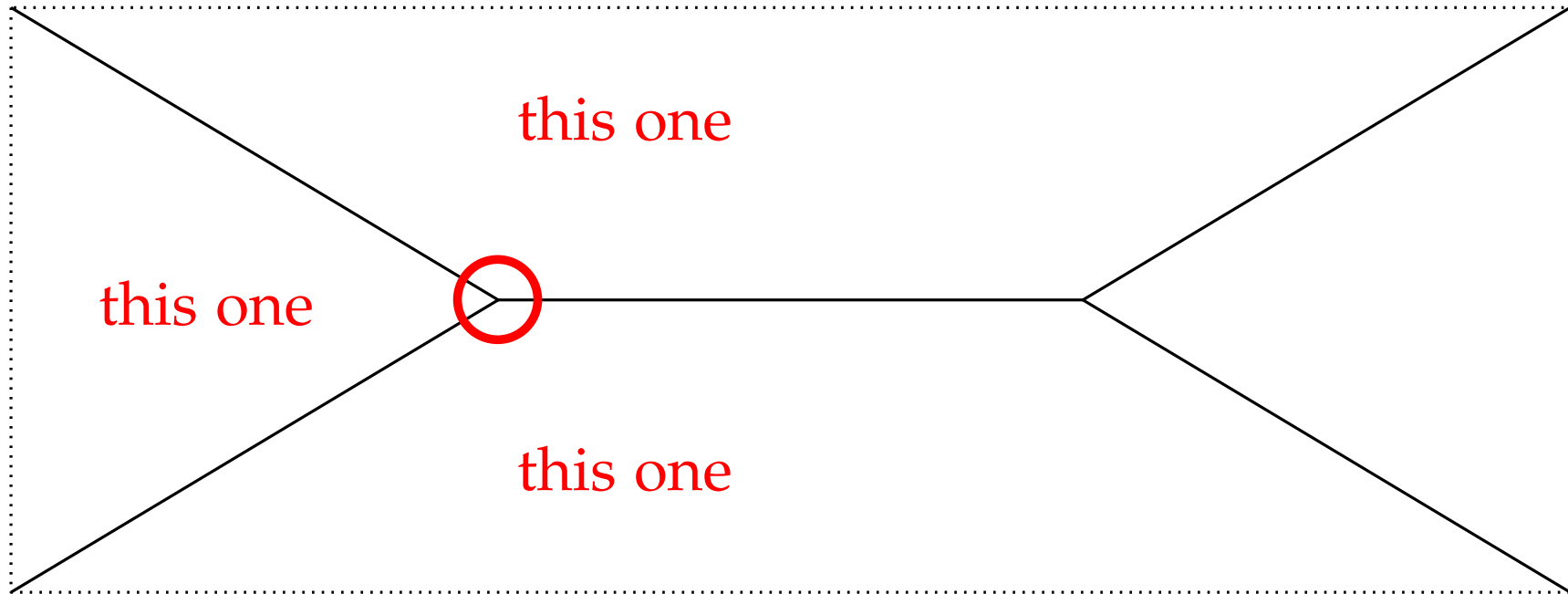
- $\Rightarrow$ each CPF/BFS is defined by exactly one basis, rather than many
- $\Rightarrow$ CPFs with $(n - 1)$ common constraint boundaries are adjacent, i.e., the two sets of nonbasic variables differ in **exactly one** variable

Adjacency (special case)



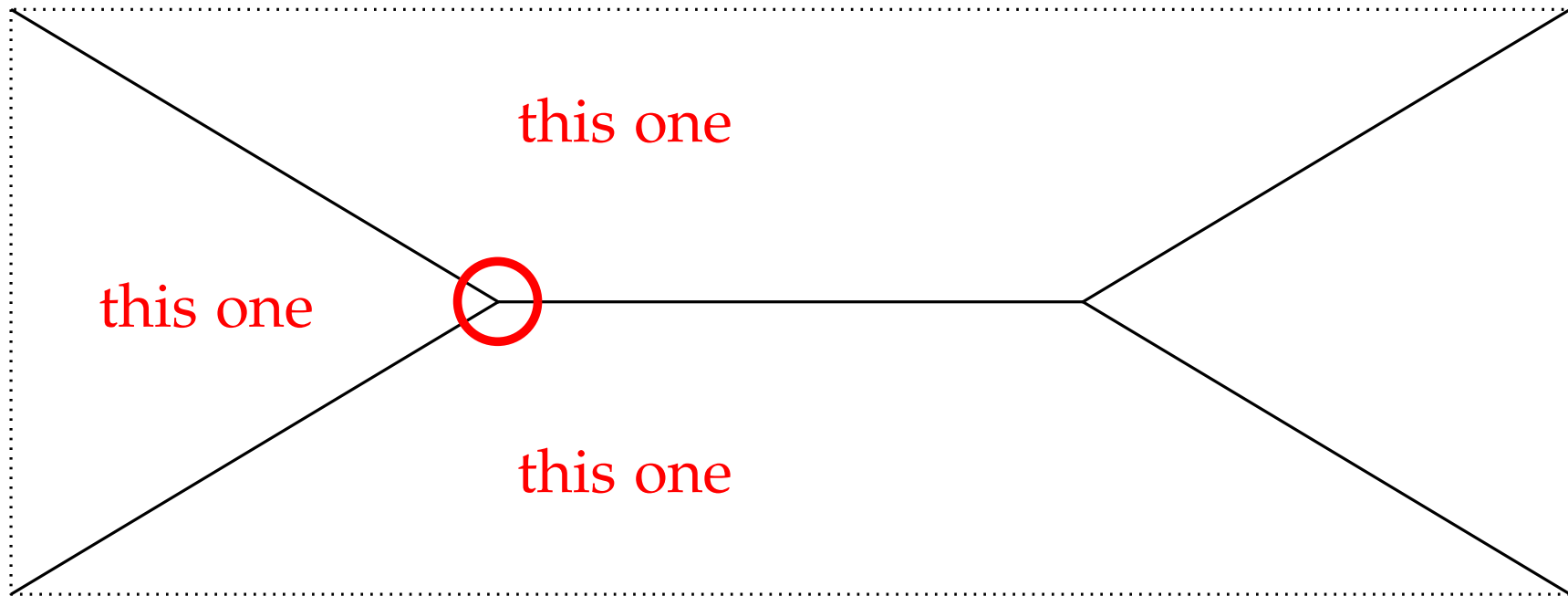
Imagine this is a 3-dimensional feasible region where the horizontal line in the middle is sticking furthest out of the page (i.e. is closest to you)

Adjacency (special case)



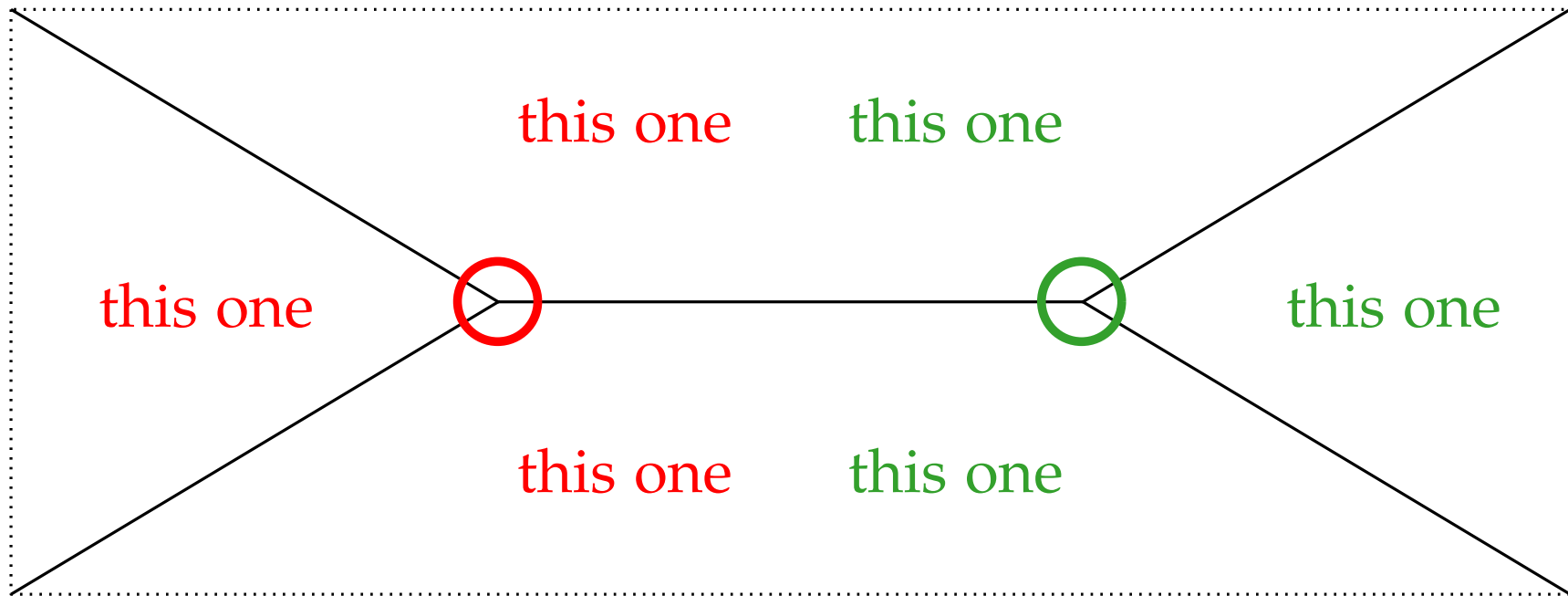
The circled CPF/BFS is defined by three constraint boundaries

Adjacency (special case)



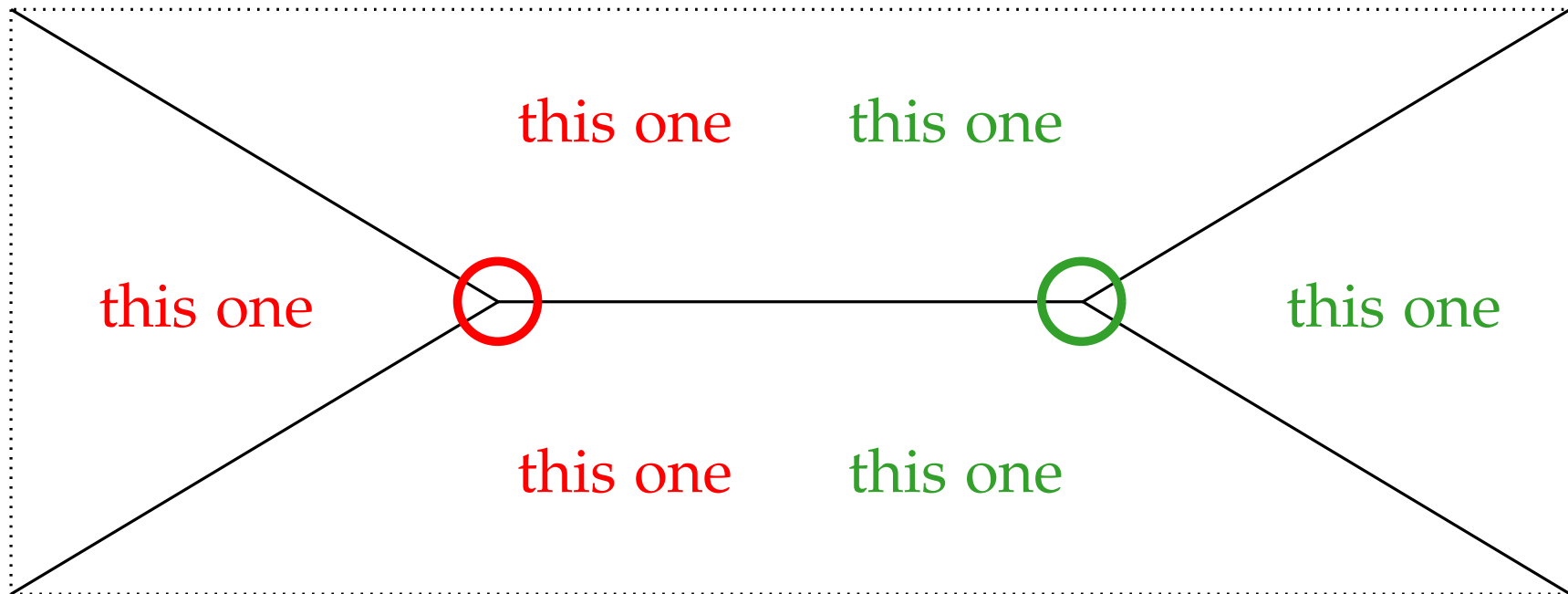
The circled CPF/BFS is defined by three constraint boundaries
their slack variables can be considered **nonbasic**

Adjacency (special case)



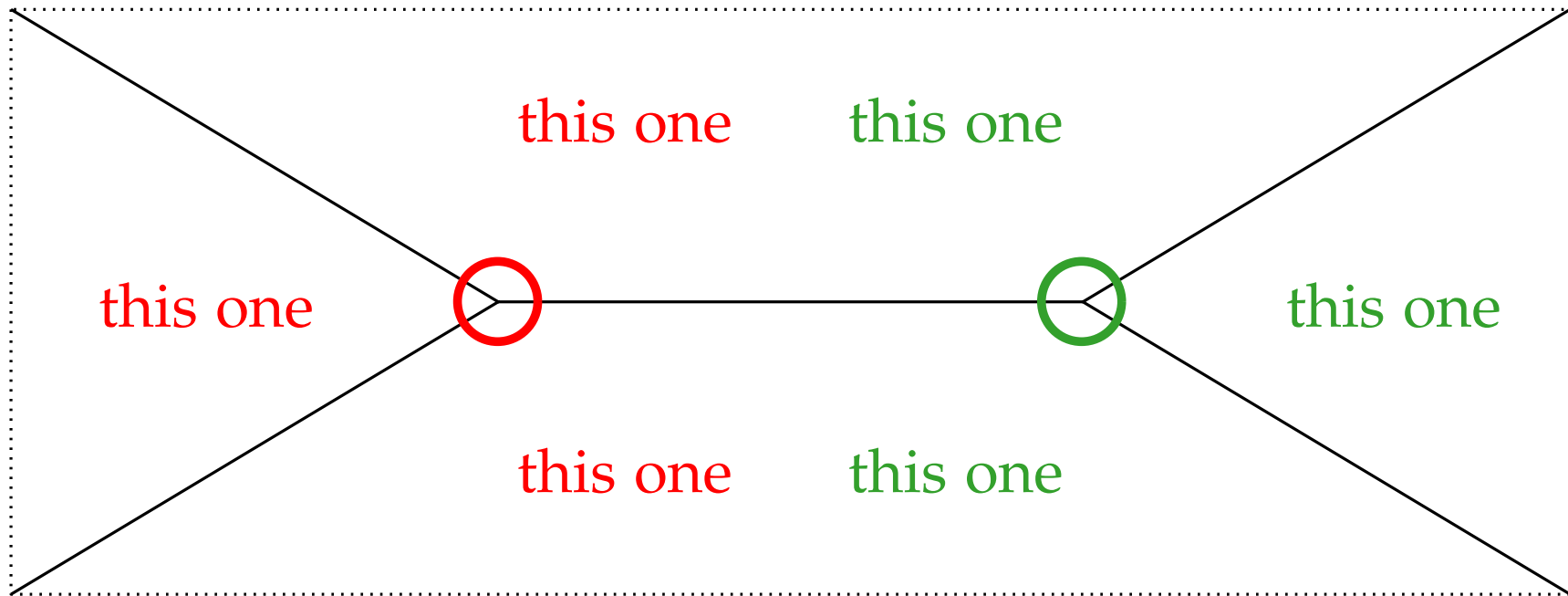
Consider the new CPF/BFS, shown in green and its constraints.

Adjacency (special case)



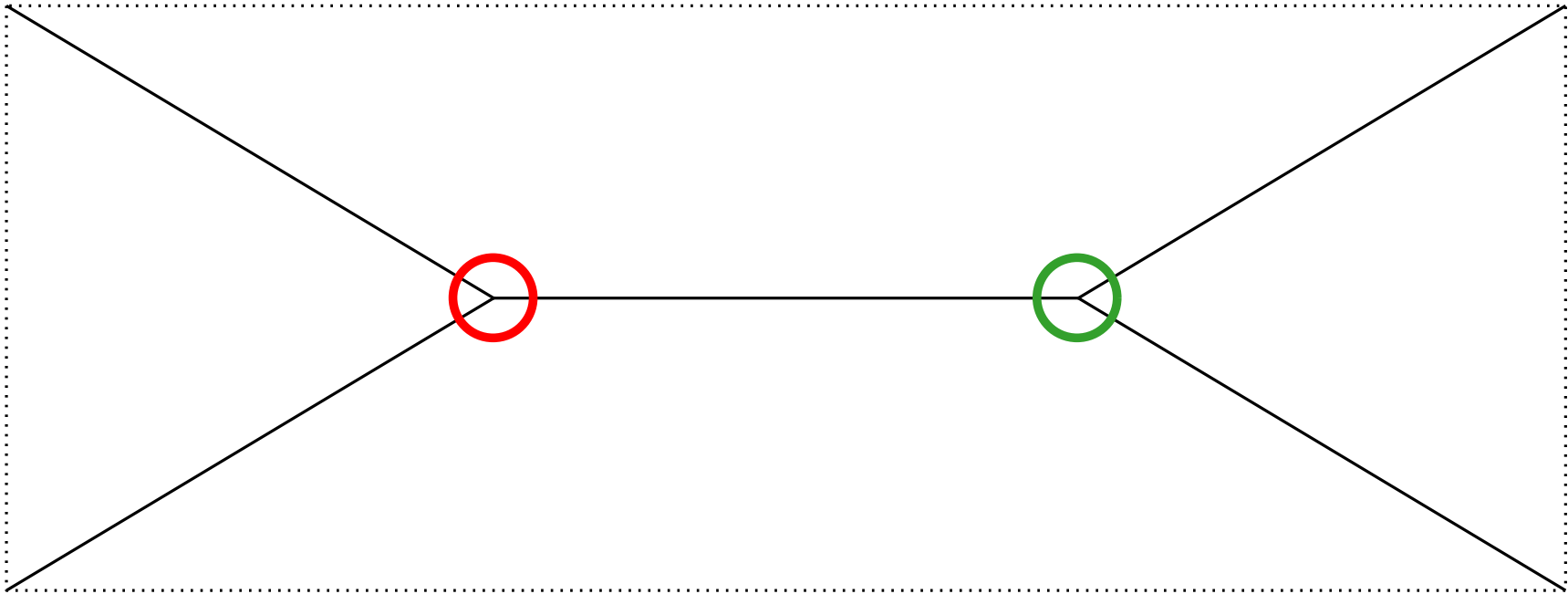
Consider the new CPF/BFS, shown in green and its constraints. The slack variables of its constraints can be considered **nonbasic**

Adjacency (special case)



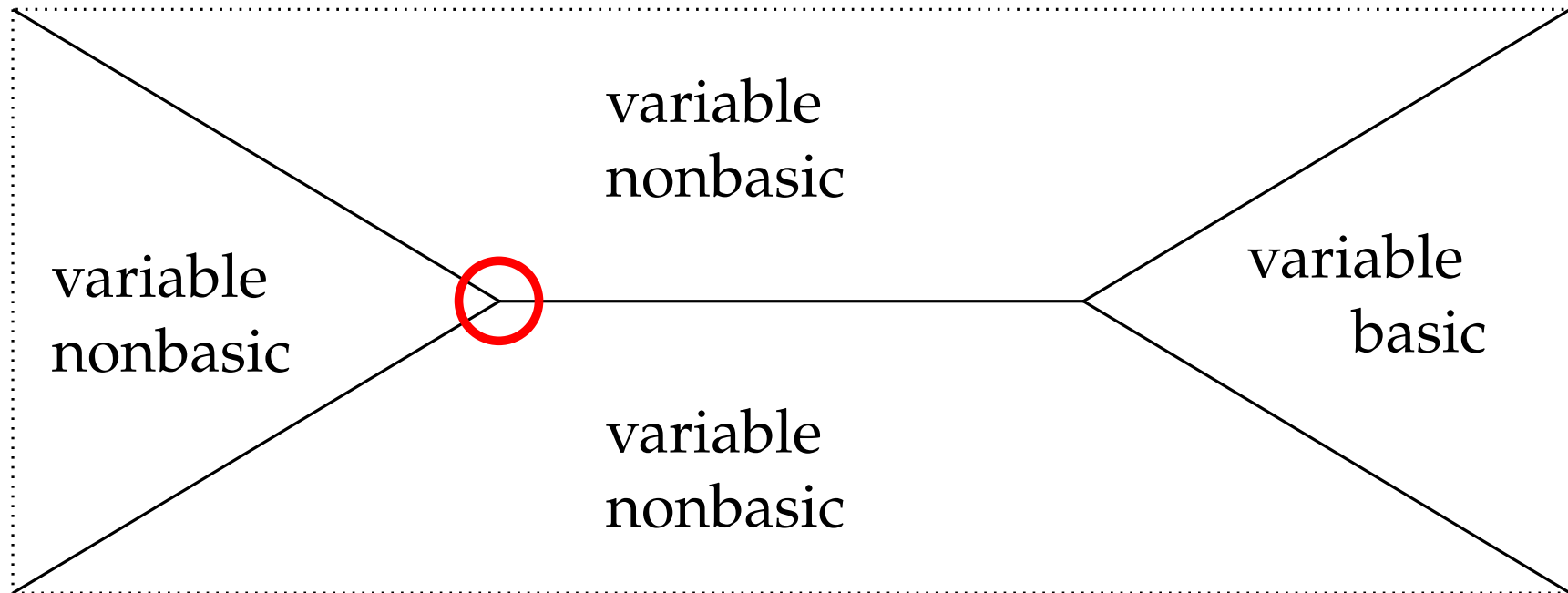
They share $n - 1 = 2$ constraint boundaries and are adjacent!
 $n - 1$ common boundaries $\Rightarrow$ the two adjacent CPFs/BFSs are connected by a line (an *edge* of the feasible region) i.e. a 1-dimensional structure

Adjacency (special case)



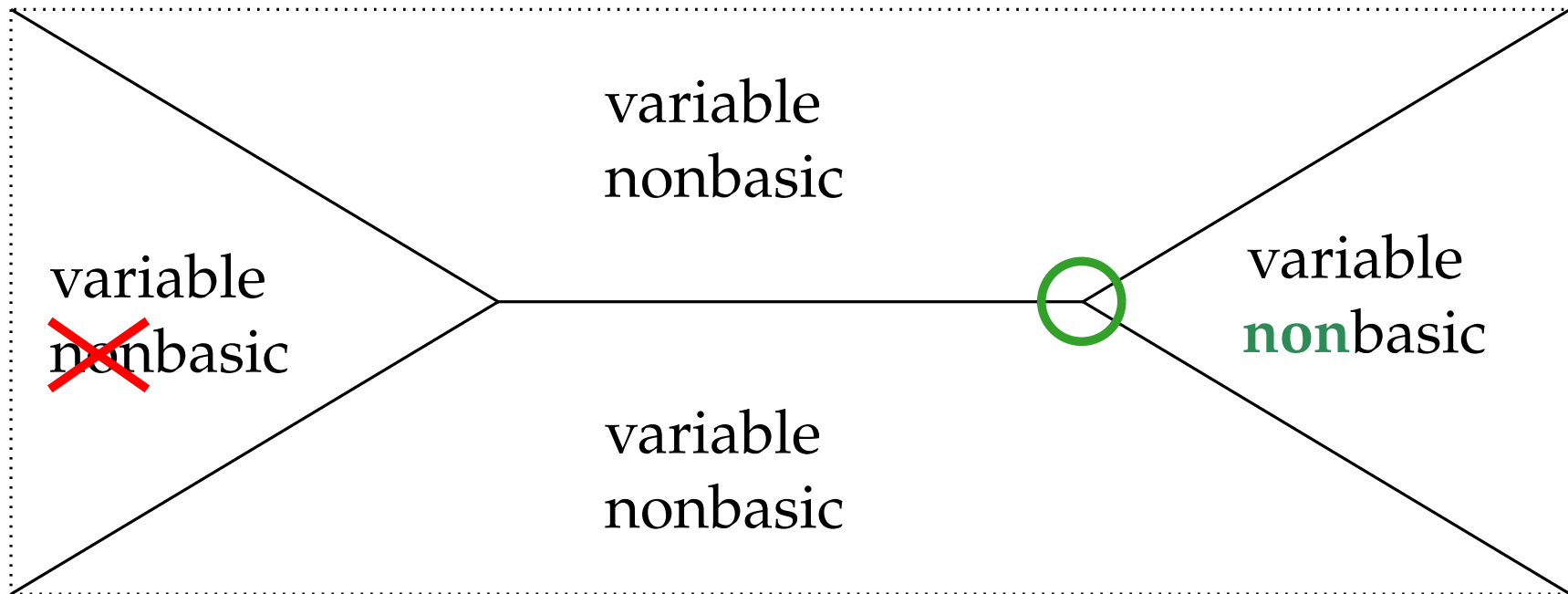
To go from red to green we:

Adjacency (special case)



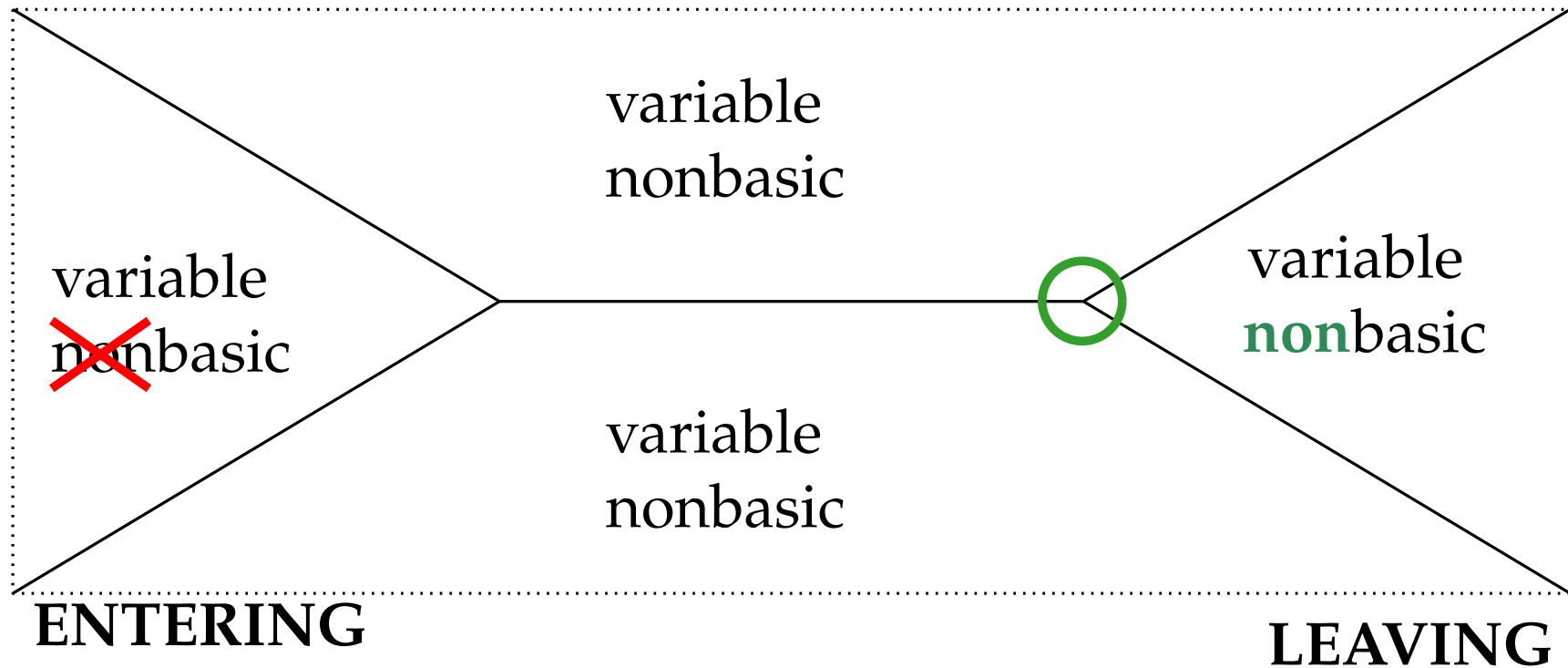
To go from red to green we:
start from this

Adjacency (special case)



To go from red to green we:
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The Simplex Method

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The Simplex Method

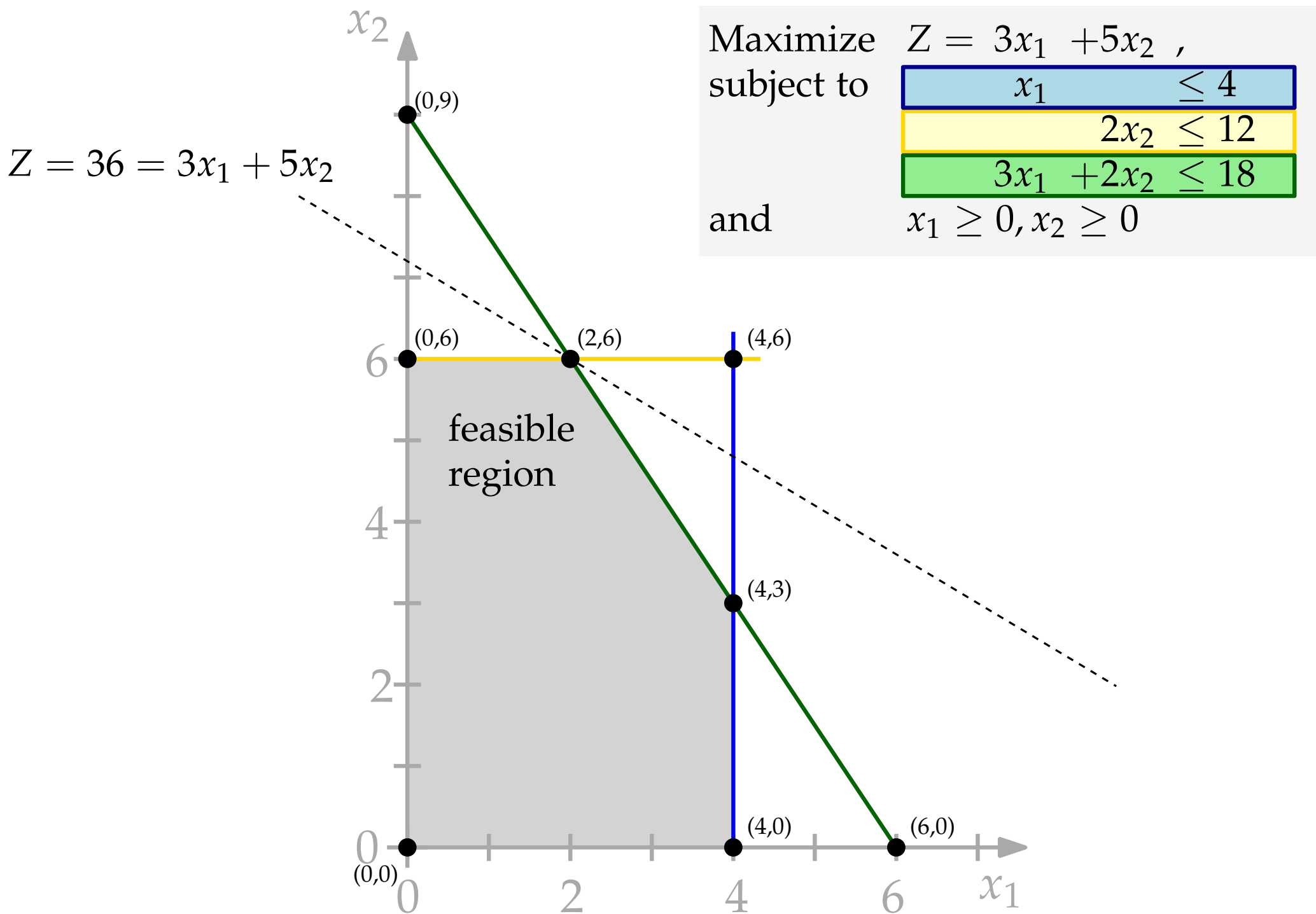
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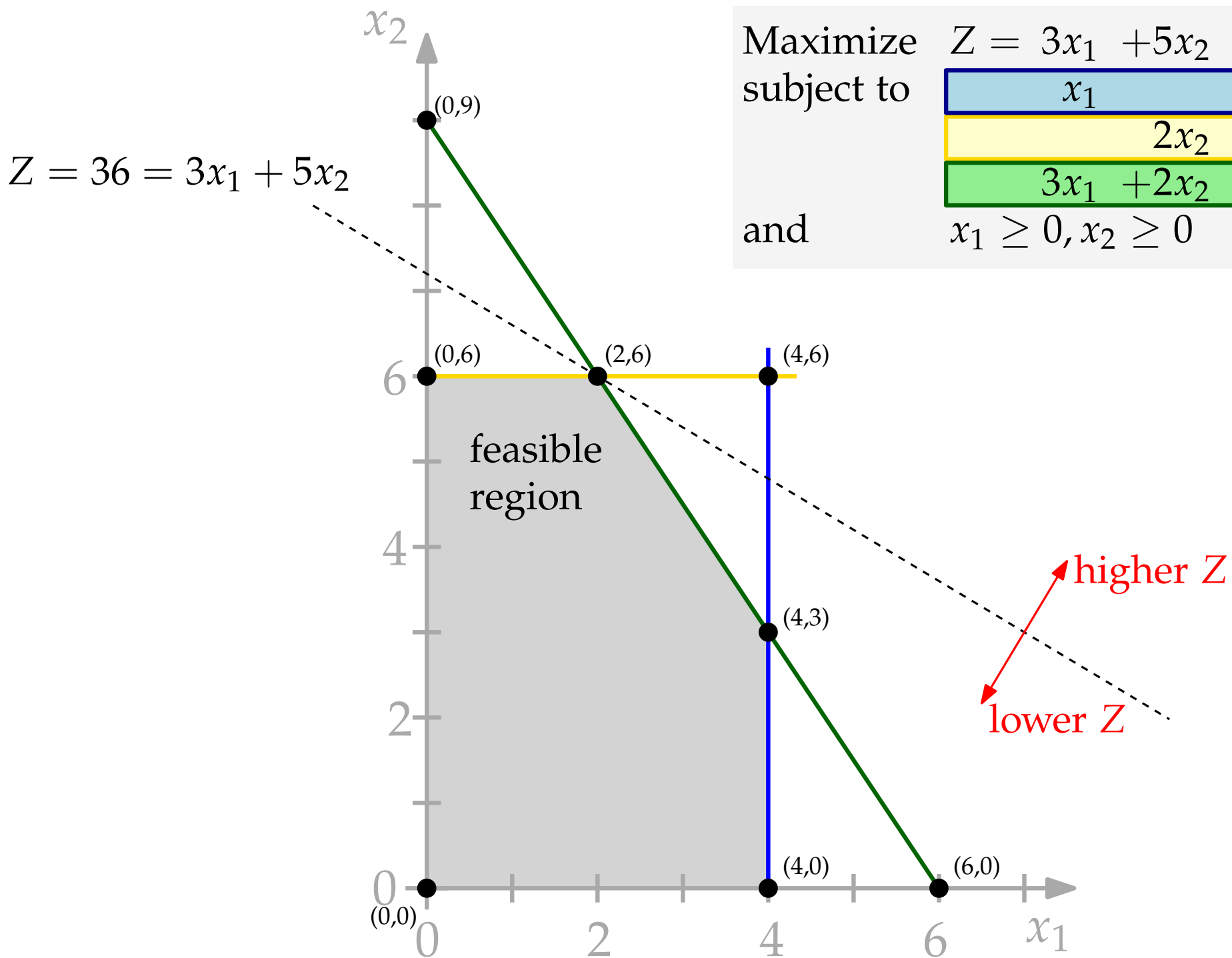
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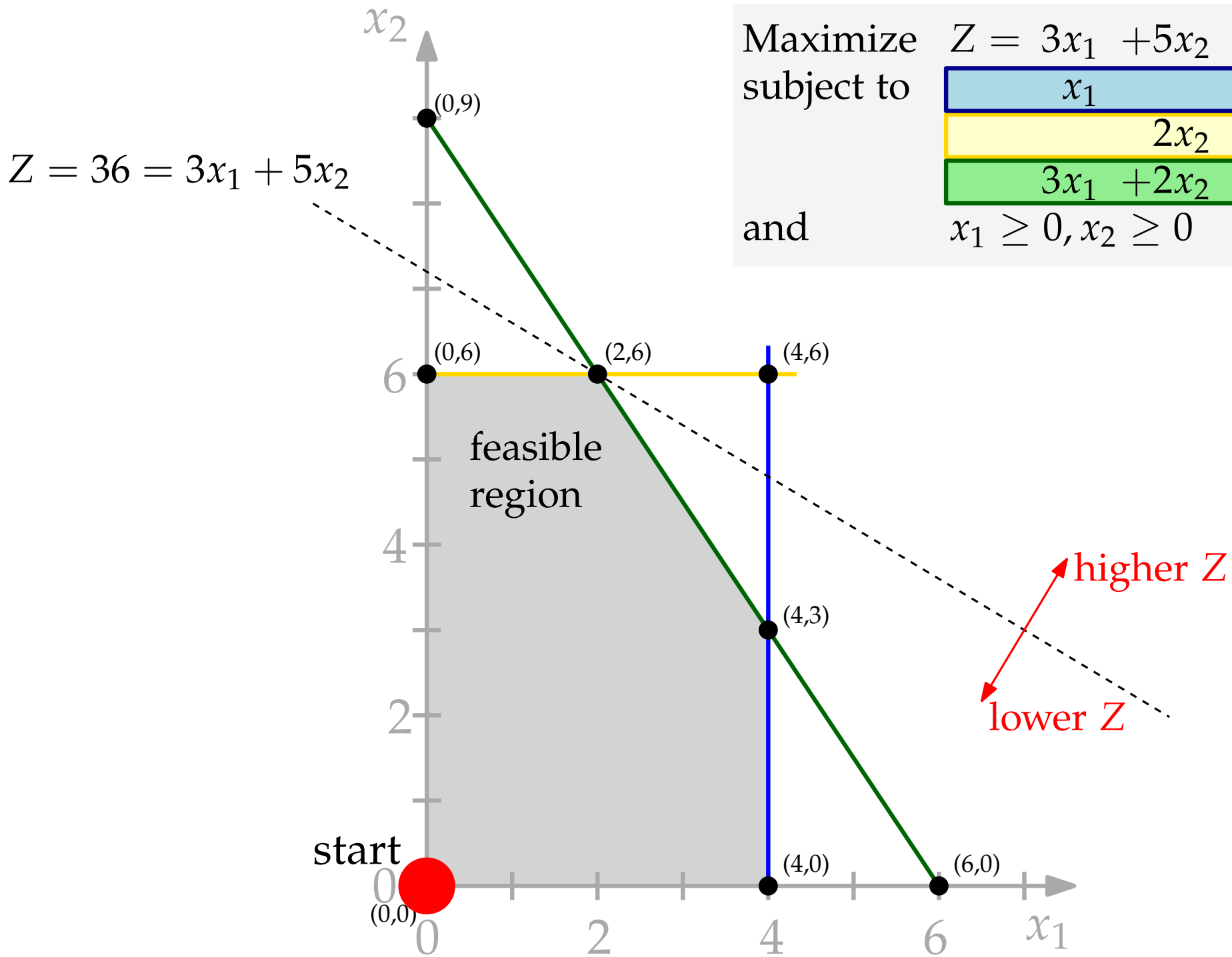
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- Iteratively move from a CPF/BFS to an adjacent CPF/BFS.
- It is *greedy*: moves to a neighbouring CPF/BFS with the most potential to improve the objective function. (It never chooses neighbouring CPF/BFS that worsens the objective function).

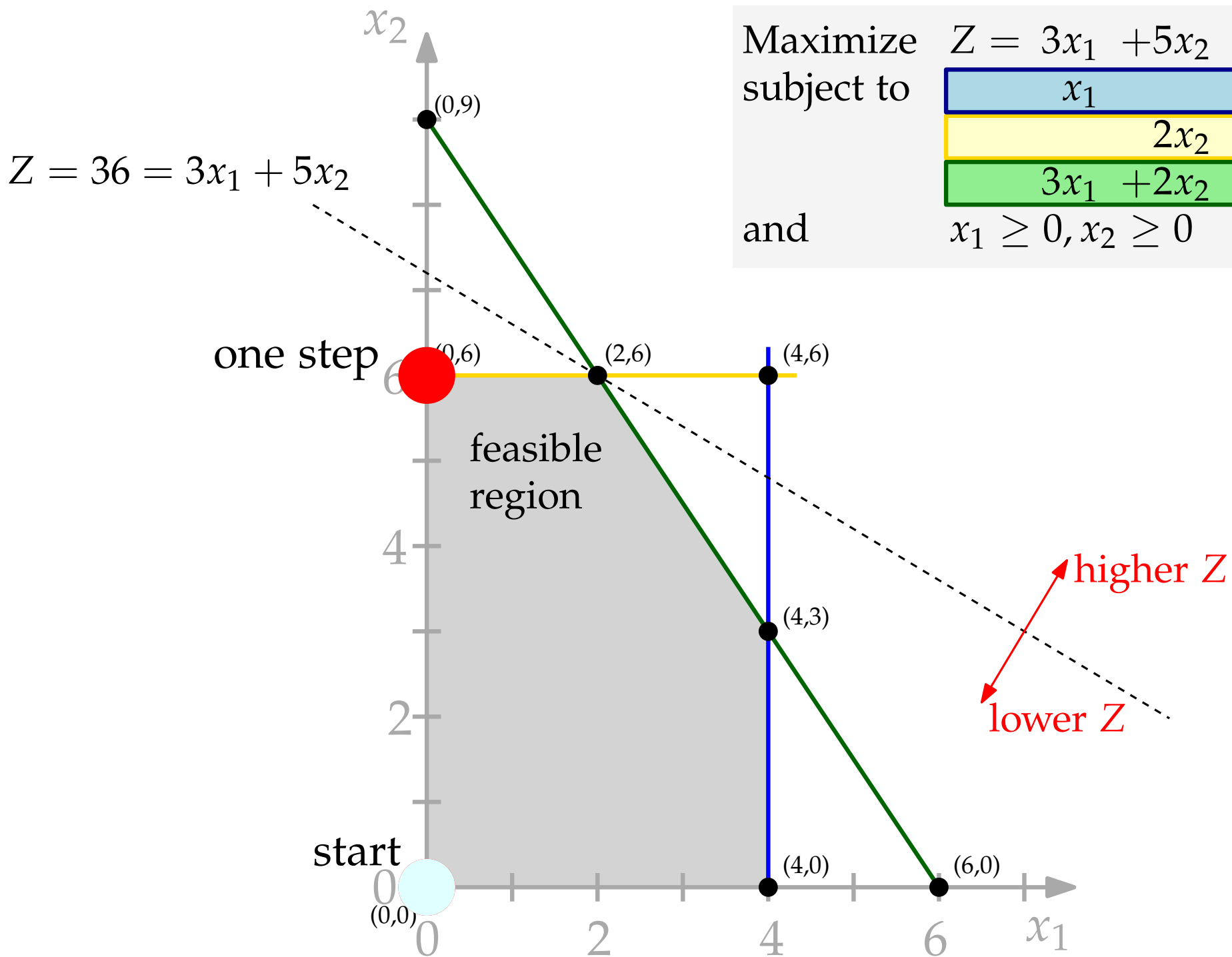
The Simplex Method

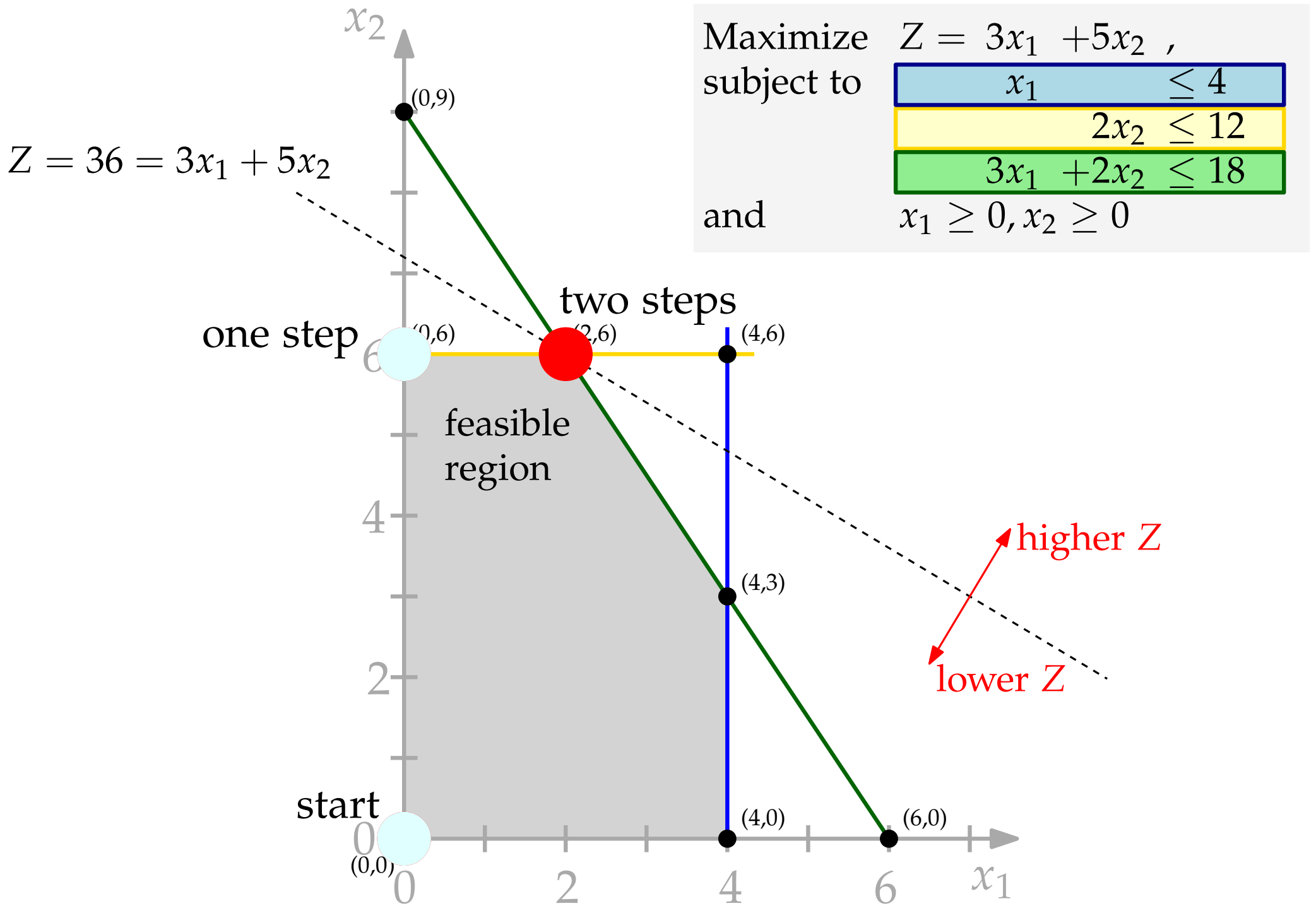
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- Stop when no improvement is available on adjacent solutions.

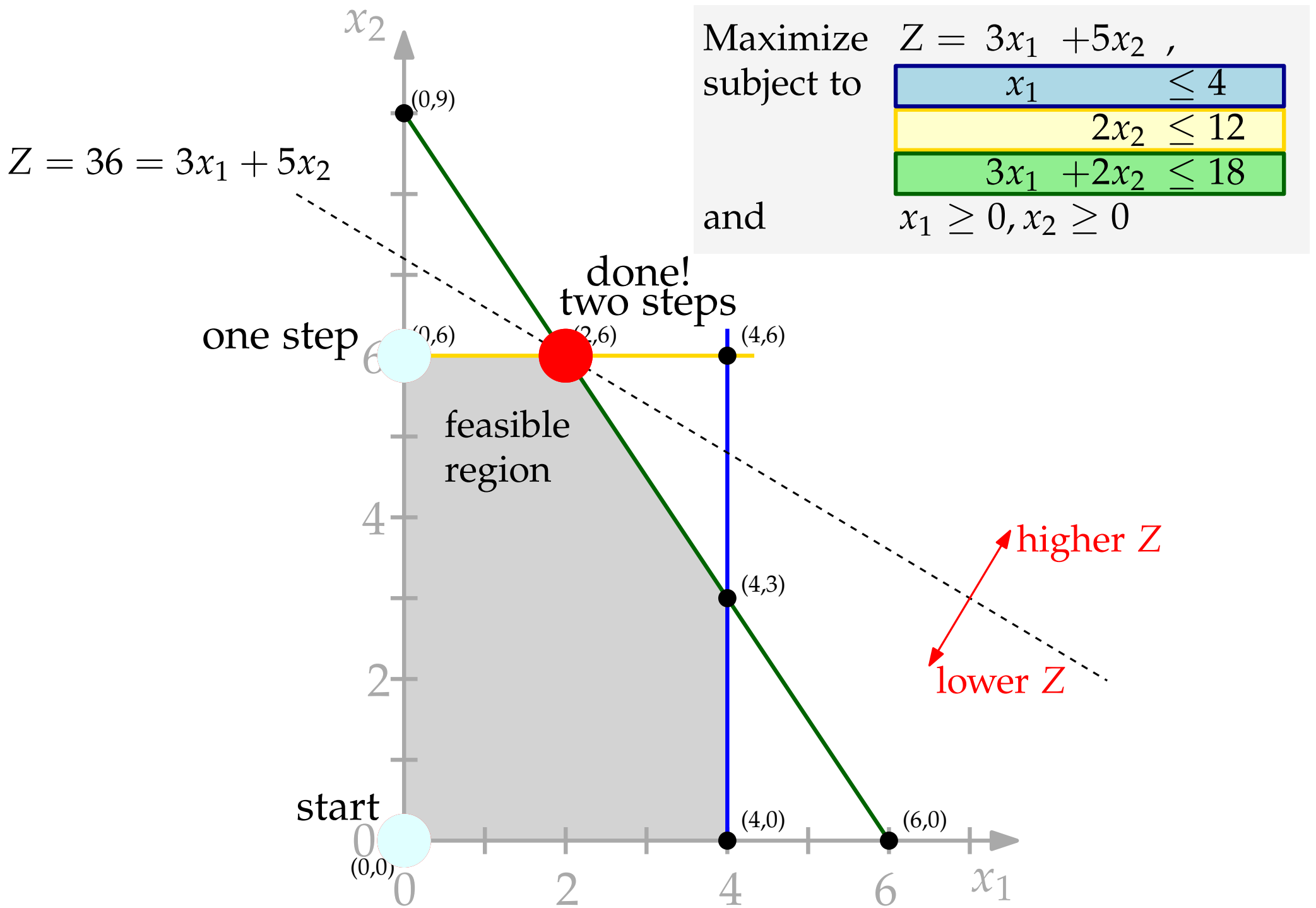












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How do you know when you've reached optimality?

- if the LP is feasible and has a bounded optimum, there exists at least one optimal CPF/BFS.
- if none of the neighbours of the current CPF/BFS can improve the objective function, we have reached optimality.
 $\Rightarrow$ suffices to look at adjacent CPF/BFS. This is an immensely powerful feature of the Simplex algorithm.

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- **Pivoting Rule**
- Move in a direction with the most **per unit** increase in the objective function.

Geometrically: the **steepest angle** of approach to the objective function.

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How do we ensure that our Basis really corresponds to linear independent columns?

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- Machinery of the Simplex Method

Simplex Tableau

Maximize Z , subject to

$$\begin{array}{llll}
 (0) & Z - 3x_1 - 5x_2 & & = 0 \\
 (1) & x_1 & + x_3 & = 4 \\
 (2) & & 2x_2 + x_4 & = 12 \\
 (3) & 3x_1 + 2x_2 & & + x_5 = 18
 \end{array}$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

For convenience, we treat the current value of the objective function (Z) as a variable

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For convenience, we treat the current value of the objective function (Z) as a variable
 $\Rightarrow$ at every iteration, the objective value occurs on the RHS, along with the values of the basic variables.

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Maximize Z , subject to

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and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

- Start with $\{x_3, x_4, x_5\}$ as the basis, all other variables as nonbasic, i.e., $x_1 = x_2 = 0$. Thus, $x_3 = 4$, $x_4 = 12$, $x_5 = 18$, and $Z = 0$.

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- Note: NO basic variables in the objective function! (trivial at the start, but Simplex will ensure this at EVERY step)

$\Rightarrow$ if we allow a **single nonbasic** variable to become basic, we can directly see the improvement in Z **per unit** increase of that variable

Simplex Tableau

Maximize Z , subject to

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 (0) & Z - 3x_1 & -5x_2 & & = 0 \\
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 (3) & 3x_1 & +2x_2 & & +x_5 = 18
 \end{array}$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

- Start with $\{x_3, x_4, x_5\}$ as the basis, all other variables as nonbasic, i.e., $x_1 = x_2 = 0$. Thus, $x_3 = 4$, $x_4 = 12$, $x_5 = 18$, and $Z = 0$.
- Note: NO basic variables in the objective function! (trivial at the start, but Simplex will ensure this at EVERY step)

$\Rightarrow$ if we allow a **single nonbasic** variable to become basic, we can directly see the improvement in Z **per unit** increase of that variable

E.g. Change in Z per unit increase x_1 : 3
 Change in Z per unit increase x_2 : 5

Simplex Tableau

Maximize Z , subject to

$$\begin{array}{rclcl}
 (0) & Z - 3x_1 & -5x_2 & & = 0 \\
 (1) & x_1 & & +x_3 & = 4 \\
 (2) & & 2x_2 & +x_4 & = 12 \\
 (3) & 3x_1 & +2x_2 & & +x_5 = 18
 \end{array}$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

- Start with $\{x_3, x_4, x_5\}$ as the basis, all other variables as nonbasic, i.e., $x_1 = x_2 = 0$. Thus, $x_3 = 4$, $x_4 = 12$, $x_5 = 18$, and $Z = 0$.
- Note: NO basic variables in the objective function! (trivial at the start, but Simplex will ensure this at EVERY step)

$\Rightarrow$ if we allow a **single nonbasic** variable to become basic, we can directly see the improvement in Z **per unit** increase of that variable

E.g. $\left. \begin{array}{l} \text{Change in } Z \text{ per unit increase } x_1 : 3 \\ \text{Change in } Z \text{ per unit increase } x_2 : 5 \end{array} \right\} \text{ By our pivoting rule, } x_2 \text{ will } \mathbf{enter}$

Simplex Tableau

Maximize Z , subject to

$$\begin{array}{rclcl}
 (0) & Z - 3x_1 & -5x_2 & & = 0 \\
 (1) & x_1 & & +x_3 & = 4 \\
 (2) & & 2x_2 & +x_4 & = 12 \\
 (3) & 3x_1 & +2x_2 & & +x_5 = 18
 \end{array}$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

Increase x_2 (decrease basic variables) until one hits zero.
(note: all other nonbasics stay at 0)

- Start with $\{x_3, x_4, x_5\}$ as the basis, all other variables as nonbasic, i.e., $x_1 = x_2 = 0$. Thus, $x_3 = 4$, $x_4 = 12$, $x_5 = 18$, and $Z = 0$.
- Note: NO basic variables in the objective function! (trivial at the start, but Simplex will ensure this at EVERY step)

$\Rightarrow$ if we allow a **single nonbasic** variable to become basic, we can directly see the improvement in Z **per unit** increase of that variable

E.g. $\left. \begin{array}{l} \text{Change in } Z \text{ per unit increase } x_1 : 3 \\ \text{Change in } Z \text{ per unit increase } x_2 : 5 \end{array} \right\} \text{ By our pivoting rule, } x_2 \text{ will } \mathbf{enter}$

Simplex Tableau

Maximize Z , subject to

$$\begin{array}{rclcl}
 (0) & Z - 3x_1 & -5x_2 & & = 0 \\
 (1) & x_1 & & +x_3 & = 4 \\
 (2) & & 2x_2 & +x_4 & = 12 \\
 (3) & 3x_1 & +2x_2 & & +x_5 = 18
 \end{array}$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

Increase x_2 (decrease basic variables) until one hits zero.
(note: all other nonbasics stay at 0)

$$\begin{array}{l}
 x_3 = 4 \\
 x_4 = 12 - 2x_2 \\
 x_5 = 18 - 2x_2
 \end{array}$$

Simplex Tableau

Maximize Z , subject to

$$\begin{array}{rclcl}
 (0) & Z - 3x_1 & -5x_2 & & = 0 \\
 (1) & x_1 & & +x_3 & = 4 \\
 (2) & & 2x_2 & +x_4 & = 12 \\
 (3) & 3x_1 & +2x_2 & & +x_5 = 18
 \end{array}$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

Increase x_2 (decrease basic variables) until one hits zero.
(note: all other nonbasics stay at 0)

$$\begin{array}{l}
 x_3 = 4 \\
 x_4 = 12 - 2x_2 \\
 x_5 = 18 - 2x_2
 \end{array}$$

- Note that x_3 is unchanged by an increase in x_2 , because the constraint boundary it represents, the vertical line at $x_3 = 4$, is **parallel** to the direction of increase.

Simplex Tableau

Maximize Z , subject to

$$\begin{array}{rclcl}
 (0) & Z - 3x_1 & -5x_2 & & = 0 \\
 (1) & x_1 & & +x_3 & = 4 \\
 (2) & & 2x_2 & +x_4 & = 12 \\
 (3) & 3x_1 & +2x_2 & & +x_5 = 18
 \end{array}$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

Increase x_2 (decrease basic variables) until one hits zero.
(note: all other nonbasics stay at 0)

$$\begin{array}{l}
 x_3 = 4 \\
 x_4 = 12 - 2x_2 \\
 x_5 = 18 - 2x_2
 \end{array}$$

- Note that x_3 is unchanged by an increase in x_2 , because the constraint boundary it represents, the vertical line at $x_3 = 4$, is **parallel** to the direction of increase.
- Note: disregard rows with non-positive (≤ 0) coefficients in the pivot column

Simplex Tableau

Maximize Z , subject to

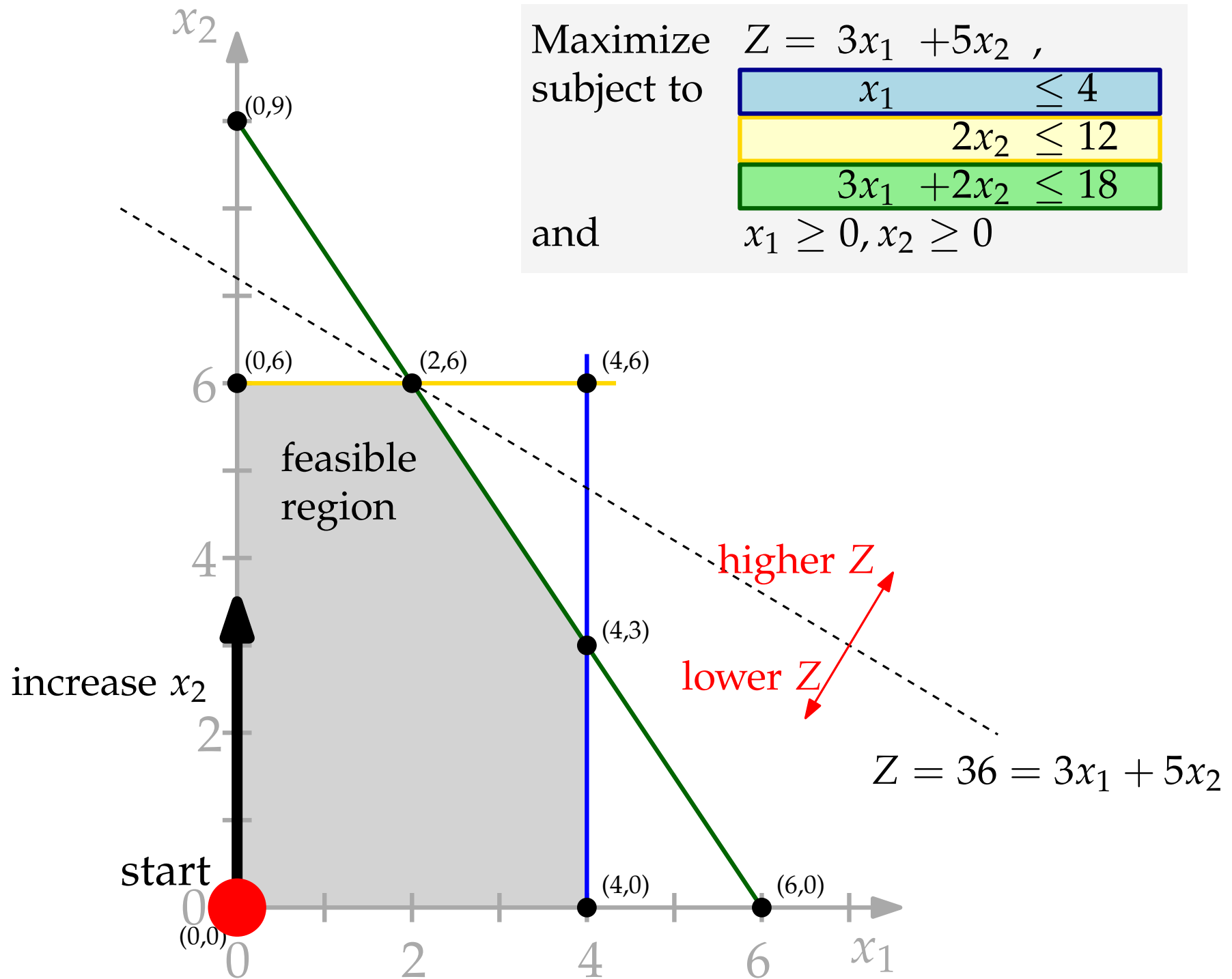
$$\begin{array}{rclcl}
 (0) & Z - 3x_1 & -5x_2 & & = 0 \\
 (1) & x_1 & & +x_3 & = 4 \\
 (2) & & 2x_2 & +x_4 & = 12 \\
 (3) & 3x_1 & +2x_2 & & +x_5 = 18
 \end{array}$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

Increase x_2 (decrease basic variables) until one hits zero.
(note: all other nonbasics stay at 0)

$$\begin{array}{l}
 x_3 = 4 \\
 x_4 = 12 - 2x_2 \\
 x_5 = 18 - 2x_2
 \end{array}$$

- Note that x_3 is unchanged by an increase in x_2 , because the constraint boundary it represents, the vertical line at $x_3 = 4$, is **parallel** to the direction of increase.
- Note: disregard rows with non-positive (≤ 0) coefficients in the pivot column
- Since $12/2 < 18/2$, the first basic variable that hits 0 is x_4 !
This is the **MINIMUM RATIO TEST**.



Maximize $Z = 3x_1 + 5x_2$,
 subject to

$$x_1 \leq 4$$

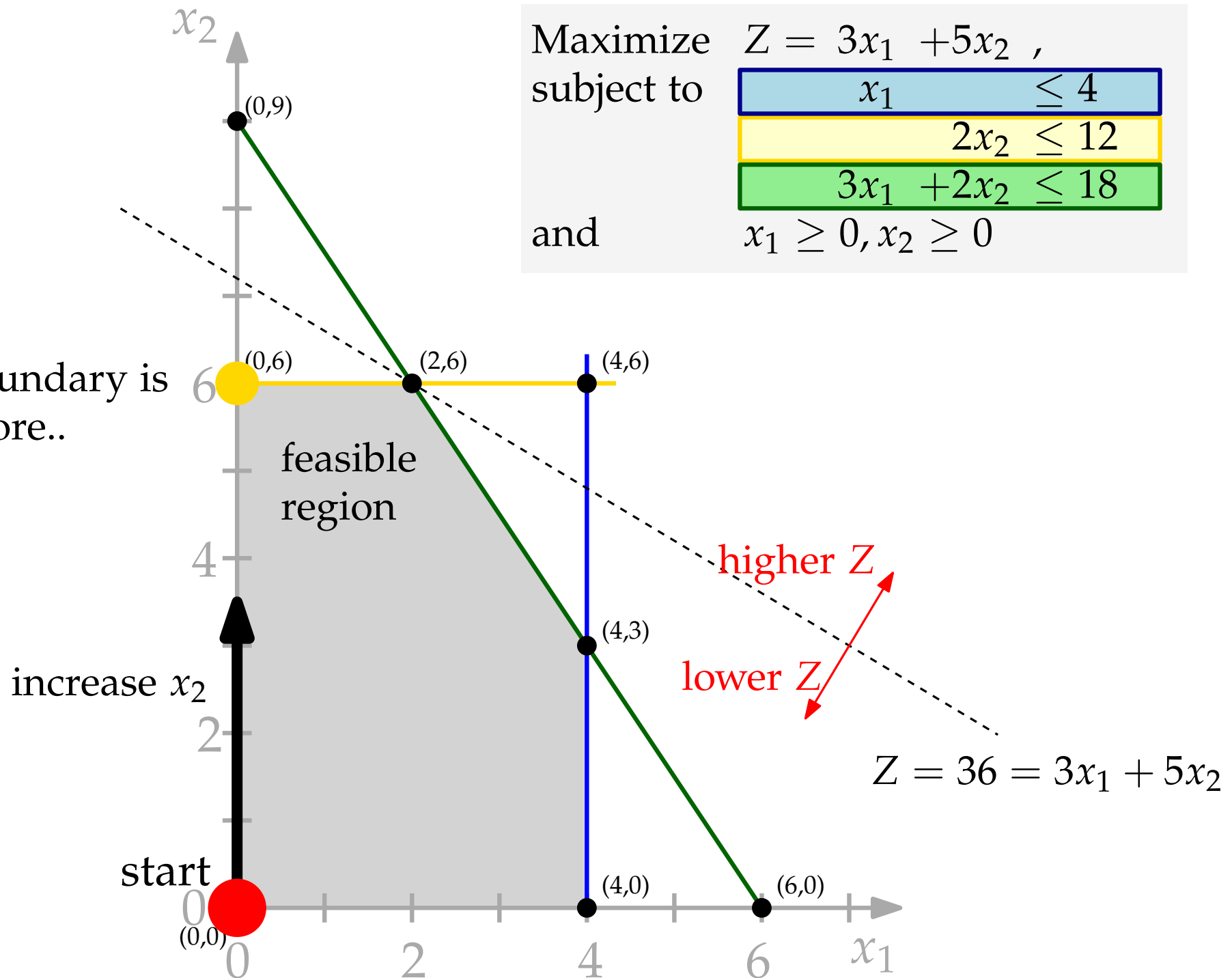
$$2x_2 \leq 12$$

$$3x_1 + 2x_2 \leq 18$$

and

$$x_1 \geq 0, x_2 \geq 0$$

this boundary is
hit before..



Maximize $Z = 3x_1 + 5x_2$,
 subject to

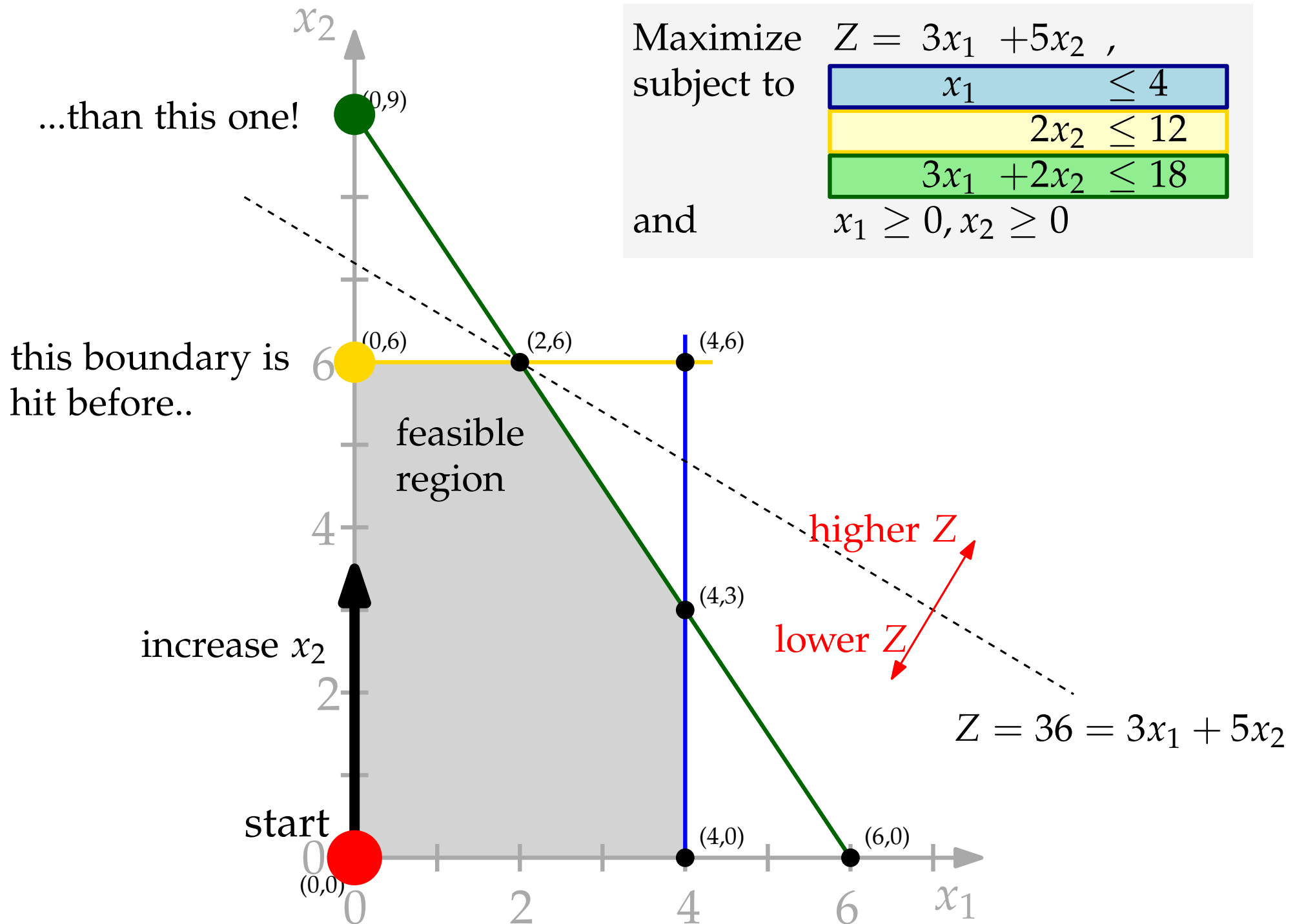
$$x_1 \leq 4$$

$$2x_2 \leq 12$$

$$3x_1 + 2x_2 \leq 18$$

and

$$x_1 \geq 0, x_2 \geq 0$$



Simplex Tableau

Maximize Z , subject to

$$\begin{array}{rclcl}
 (0) & Z - 3x_1 & -5x_2 & & = 0 \\
 (1) & x_1 & & +x_3 & = 4 \\
 (2) & & 2x_2 & +x_4 & = 12 \\
 (3) & 3x_1 & +2x_2 & & +x_5 = 18
 \end{array}$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

(note: all other nonbasics stay at 0)

Increase x_2 (lowering basic variables) until one hits zero.

$$\begin{array}{l}
 x_3 = 4 \\
 x_4 = 12 - 2x_2 \\
 x_5 = 18 - 2x_2
 \end{array}$$

- Note that x_3 is unchanged by an increase in x_2 , because the constraint boundary it represents, the vertical line at $x_1 = 4$, is **parallel** to the direction of increase.
- Since $12/2 < 18/2$, the first basic variable that hits 0 is x_4 !
This is the **MINIMUM RATIO TEST**.

This ensures we stay within the feasible region.

Simplex Tableau

Maximize Z , subject to

$$\begin{array}{rclcl}
 (0) & Z - 3x_1 & -5x_2 & & = 0 \\
 (1) & x_1 & & +x_3 & = 4 \\
 (2) & & 2x_2 & +x_4 & = 12 \\
 (3) & 3x_1 & +2x_2 & & +x_5 = 18
 \end{array}$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

To sum up:

1. Pick the **entering** variable by the coefficient in the objective (here: x_2)
2. Pick who **leaves** by the Minimum Ratio Test. (here: x_4)

Simplex Tableau

Maximize Z , subject to

$$(0) \quad Z - 3x_1 - 5x_2 = 0$$

$$(1) \quad x_1 + x_3 = 4$$

$$(2) \quad 2x_2 + x_4 = 12$$

$$(3) \quad 3x_1 + 2x_2 + x_5 = 18$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

Now update the Tableau according to this change!

Simplex Tableau

Maximize Z , subject to

$$\begin{array}{rclcl}
 (0) & Z - 3x_1 & -5x_2 & & = 0 \\
 (1) & x_1 & & +x_3 & = 4 \\
 (2) & & 2x_2 & +x_4 & = 12 \\
 (3) & 3x_1 & +2x_2 & & +x_5 = 18
 \end{array}$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

Now update the Tableau according to this change!

- update x_2 's column so that it has all 0s except for one **1** in the row of “owned” by the leaving variable x_4 .

Simplex Tableau

Maximize Z , subject to

$$\begin{array}{rclclcl}
 (0) & Z - 3x_1 & -5x_2 & & & = 0 \\
 (1) & x_1 & & +x_3 & & = 4 \\
 (2) & & 2x_2 & & +x_4 & = 12 \\
 (3) & 3x_1 & +2x_2 & & & +x_5 = 18
 \end{array}$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

Now update the Tableau according to this change!

- update x_2 's column so that it has all 0s except for one **1** in the row of “owned” by the leaving variable x_4 .
- **eliminate** x_2 from the objective function (and other rows)

Simplex Tableau

Maximize Z , subject to

$$\begin{array}{rclclcl}
 (0) & Z - 3x_1 & -5x_2 & & & = 0 \\
 (1) & x_1 & & +x_3 & & = 4 \\
 (2) & & 2x_2 & & +x_4 & = 12 \\
 (3) & 3x_1 & +2x_2 & & & +x_5 = 18
 \end{array}$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

Now update the Tableau according to this change!

- update x_2 's column so that it has all 0s except for one **1** in the row of "owned" by the leaving variable x_4 .
- **eliminate** x_2 from the objective function (and other rows)
- use Gaussian Elimination (i.e., elementary row operations) to make these changes!

Simplex Tableau

Maximize Z , subject to

$$(0) \quad Z - 3x_1 \quad \boxed{} \quad + \frac{5}{2}x_4 \quad = 30$$

$$(1) \quad x_1 \quad \boxed{} \quad + x_3 \quad = 4$$

$$(2) \quad \quad x_2 \quad \boxed{} \quad + \frac{1}{2}x_4 \quad = 6$$

$$(3) \quad 3x_1 \quad \boxed{} \quad - x_4 \quad + x_5 \quad = 6$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

$$\text{row}(0) \leftarrow \text{row}(0) + 5 \text{ row}(2)$$

$$\begin{aligned} \text{row}(2) &\leftarrow \frac{1}{2} \text{ row}(2), \text{ then} \\ \text{row}(3) &\leftarrow \text{row}(3) - 2 \text{ row}(2) \end{aligned}$$

Now update the Tableau according to this change!

- update x_2 's column so that it has all 0s except for one **1** in the row of "owned" by the leaving variable x_4 .
- **eliminate** x_2 from the objective function (and other rows)
- use Gaussian Elimination (i.e., elementary row operations) to make these changes!

Simplex Tableau

Maximize Z , subject to

$$(0) \quad Z - 3x_1 \quad \quad \quad + \frac{5}{2}x_4 \quad \quad = 30$$

$$(1) \quad \quad x_1 \quad \quad + x_3 \quad \quad = 4$$

$$(2) \quad \quad \quad x_2 \quad \quad + \frac{1}{2}x_4 \quad \quad = 6$$

$$(3) \quad \quad 3x_1 \quad \quad - x_4 \quad + x_5 = 6$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

- New basis is $\{x_2, x_3, x_5\}$ and $(x_1, x_2, x_3, x_4, x_5) = (0, 6, 4, 0, 6)$.

Simplex Tableau

Maximize Z , subject to

$$\begin{array}{rcllclclcl}
 (0) & Z & -3x_1 & & & +\frac{5}{2}x_4 & & = & 30 \\
 (1) & & x_1 & & +x_3 & & & = & 4 \\
 (2) & & & x_2 & & +\frac{1}{2}x_4 & & = & 6 \\
 (3) & & 3x_1 & & & -x_4 & +x_5 & = & 6
 \end{array}$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

- New basis is $\{x_2, x_3, x_5\}$ and $(x_1, x_2, x_3, x_4, x_5) = (0, 6, 4, 0, 6)$.
- New value of objective function is 30.

Simplex Tableau

Maximize Z , subject to

$$\begin{array}{rclclclcl}
 (0) & Z & -3x_1 & & +\frac{5}{2}x_4 & & = & 30 \\
 (1) & & x_1 & & +x_3 & & = & 4 \\
 (2) & & & x_2 & & +\frac{1}{2}x_4 & = & 6 \\
 (3) & & 3x_1 & & & -x_4 & +x_5 & = & 6
 \end{array}$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

- New basis is $\{x_2, x_3, x_5\}$ and $(x_1, x_2, x_3, x_4, x_5) = (0, 6, 4, 0, 6)$.
- New value of objective function is 30.

Are we finished?

Simplex Tableau

Maximize Z , subject to

$$\begin{array}{rclclclclcl}
 (0) & Z & -3x_1 & & & +\frac{5}{2}x_4 & & = & 30 \\
 (1) & & x_1 & & +x_3 & & & = & 4 \\
 (2) & & & x_2 & & +\frac{1}{2}x_4 & & = & 6 \\
 (3) & & 3x_1 & & & -x_4 & +x_5 & = & 6
 \end{array}$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

- New basis is $\{x_2, x_3, x_5\}$ and $(x_1, x_2, x_3, x_4, x_5) = (0, 6, 4, 0, 6)$.
- New value of objective function is 30.

Are we finished?

- No! Nonbasic variable x_1 still has potential to improve the objective function. (Note: coefficient of x_4 in the top row is non-negative means that increasing x_4 does not improve the objective function.)

Simplex Tableau

Maximize Z , subject to

$$\begin{array}{rcllclcl}
 (0) & Z - 3x_1 & & + \frac{5}{2}x_4 & & = & 30 \\
 (1) & & x_1 & & + x_3 & & = 4 \\
 (2) & & & x_2 & & + \frac{1}{2}x_4 & = 6 \\
 (3) & & 3x_1 & & & - x_4 & + x_5 = 6
 \end{array}$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

- New basis is $\{x_2, x_3, x_5\}$ and $(x_1, x_2, x_3, x_4, x_5) = (0, 6, 4, 0, 6)$.
- New value of objective function is 30.

Are we finished?

- No! Nonbasic variable x_1 still has potential to improve the objective function. (Note: coefficient of x_4 in the top row is non-negative means that increasing x_4 does not improve the objective function.)
- So, x_1 enters ... who leaves?

Simplex Tableau

Maximize Z , subject to

$$(0) \quad Z - 3x_1 \quad \quad \quad + \frac{5}{2}x_4 \quad \quad = 30$$

$$(1) \quad \quad x_1 \quad \quad + x_3 \quad \quad = 4$$

$$(2) \quad \quad \quad x_2 \quad \quad + \frac{1}{2}x_4 \quad \quad = 6$$

$$(3) \quad \quad 3x_1 \quad \quad - x_4 \quad + x_5 = 6$$

$$x_3 = 4 - x_1 \rightarrow \text{zero at } x_1 = (4/1) = 4$$

x_2 : unaffected by x_1

$$x_5 = 6 - 3x_1 \rightarrow \text{zero at } x_1 = (6/3) = 2$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

- New basis is $\{x_2, x_3, x_5\}$ and $(x_1, x_2, x_3, x_4, x_5) = (0, 6, 4, 0, 6)$.
- New value of objective function is 30.

Are we finished?

- No! Nonbasic variable x_1 still has potential to improve the objective function. (Note: coefficient of x_4 in the top row is non-negative means that increasing x_4 does not improve the objective function.)
- So, x_1 enters ... who leaves?

Simplex Tableau

Maximize Z , subject to

$$(0) \quad Z - 3x_1 \quad \quad \quad + \frac{5}{2}x_4 \quad \quad = 30$$

$$(1) \quad \quad x_1 \quad \quad + x_3 \quad \quad = 4$$

$$(2) \quad \quad \quad x_2 \quad \quad + \frac{1}{2}x_4 \quad \quad = 6$$

$$(3) \quad \quad 3x_1 \quad \quad - x_4 \quad + x_5 = 6$$

$$x_3 = 4 - x_1 \rightarrow \text{zero at } x_1 = (4/1) = 4$$

x_2 : unaffected by x_1

$$x_5 = 6 - 3x_1 \rightarrow \text{zero at } x_1 = (6/3) = 2$$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

- New basis is $\{x_2, x_3, x_5\}$ and $(x_1, x_2, x_3, x_4, x_5) = (0, 6, 4, 0, 6)$.
- New value of objective function is 30.

Are we finished?

- No! Nonbasic variable x_1 still has potential to improve the objective function. (Note: coefficient of x_4 in the top row is non-negative means that increasing x_4 does not improve the objective function.)
- So, x_1 enters ... who leaves? x_5 will go!

Simplex Tableau

Maximize Z , subject to

(0)	Z			$+\frac{3}{2}x_4$	$+x_5$	$= 36$
(1)			x_3	$-\frac{1}{3}x_4$	$+\frac{1}{3}x_5$	$= 4$
(2)		x_2		$+\frac{1}{2}x_4$		$= 6$
(3)		x_1		$-\frac{1}{3}x_4$	$+\frac{1}{3}x_5$	$= 2$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

- New basis is $\{x_1, x_2, x_3\}$ and $(x_1, x_2, x_3, x_4, x_5) = (2, 6, 2, 0, 0)$.
New value of objective function is 36.

Simplex Tableau

Maximize Z , subject to

(0)	Z			$+\frac{3}{2}x_4$	$+x_5$	$= 36$
(1)			x_3	$-\frac{1}{3}x_4$	$+\frac{1}{3}x_5$	$= 4$
(2)		x_2		$+\frac{1}{2}x_4$		$= 6$
(3)		x_1		$-\frac{1}{3}x_4$	$+\frac{1}{3}x_5$	$= 2$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

- New basis is $\{x_1, x_2, x_3\}$ and $(x_1, x_2, x_3, x_4, x_5) = (2, 6, 2, 0, 0)$.
New value of objective function is 36.

Are we done?

Simplex Tableau

Maximize Z , subject to

(0)	Z			$+\frac{3}{2}x_4$	$+x_5$	$= 36$
(1)			x_3	$-\frac{1}{3}x_4$	$+\frac{1}{3}x_5$	$= 4$
(2)		x_2		$+\frac{1}{2}x_4$		$= 6$
(3)		x_1		$-\frac{1}{3}x_4$	$+\frac{1}{3}x_5$	$= 2$

and $x_j \geq 0$ for each $j \in \{1, \dots, 5\}$

- New basis is $\{x_1, x_2, x_3\}$ and $(x_1, x_2, x_3, x_4, x_5) = (2, 6, 2, 0, 0)$.
New value of objective function is 36.

Are we done?

- YES! Neither of the two nonbasic variables can improve the objective function. So we know we have reached optimality!

The Simplex Method

1. Find a BFS to start (just use the slack variables as the basis).
2. Can a currently nonbasic variable help to improve the objective function? (**just check coefficients in row(0)**)
No? We are done! We have an optimal solution!
3. Pick a nonbasic variable with the greatest potential to improve the objective function, **the entering variable**
4. Use the **minimum ratio test** to identify the **leaving variable**
5. Update the system for the new BFS using Gaussian elimination
6. Go to 2.